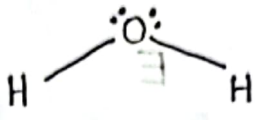
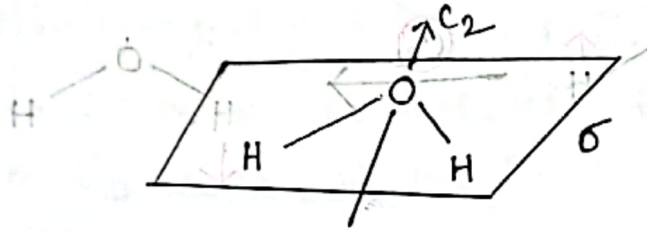
Information regarding molecules

• Water (H_2O) :

• Configuration :  [Planar]

• Symmetry elements : E, C_2, σ, σ'

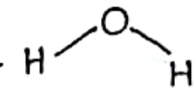


C_2 → O এর ভেতর দিয়ে principle rotation axis

σ → As, molecular plane contains the principl rotation axis, rather than being perpendicular, it is vertical plane

σ' → It is perpendicular to the molecular plane, placed at O. It's a vertical plane.

• Point group :



Does this belong to any special point group?

NO

C_n = ?
yes, n = 2

$2C_2 \perp C_2$?
NO.

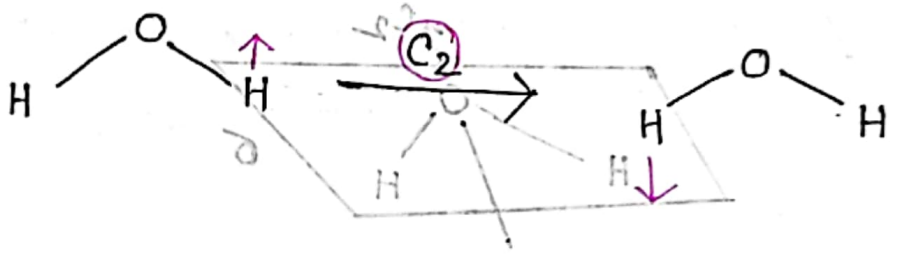
σ_h ?
NO

$2\sigma_v$?
yes → C_2V

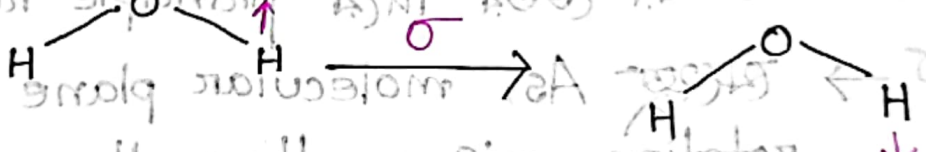
• Multiplication Table :

	E	C_2	σ	σ'
E	E	C_2	σ	σ'
C_2	C_2	E	σ'	σ
σ	σ	σ'	E	C_2
σ'	σ'	σ	C_2	E

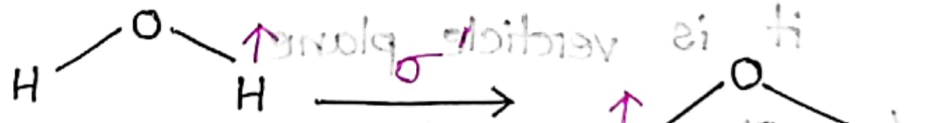
C_2 :



σ :

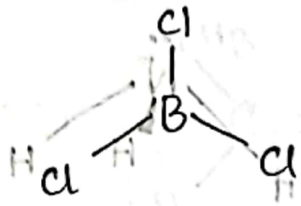


σ' :



- Boron trichloride : [BCl_3] [BF_3]

- Configuration :



[Planar]

- Symmetry Elements : E, C_3, $3C_2$, σ_h, $3\sigma_v$

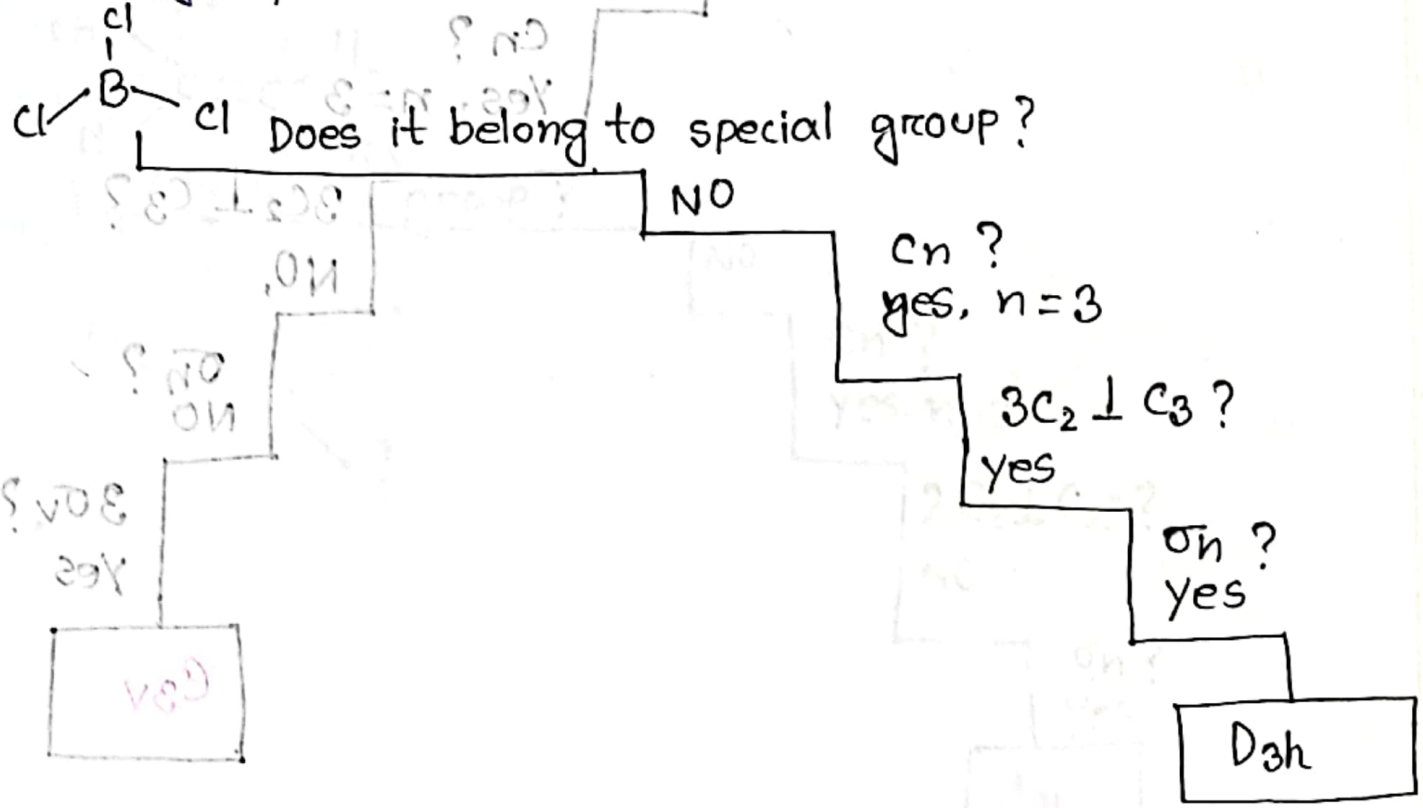
C_3 : If BCl_3 is placed in a paper and an axis passe through the B up side down, that axis is C_3.

C_2 : At every B-Cl bond, there is an C_2 axis. Three B-Cl bond represents three C_2 axis. pependicular to C_3 axis.

σ_h : As, C_3 axis is perpendicular to molecular plane, it contains horizontal plane.

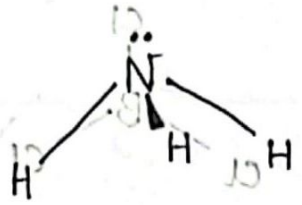
σ_v : Three vertical plane along three B-Cl bond.

- Point group :



• Ammonia [NH_3] :

Configuration :



[Non- planar]

Symmetry Elements : E, C_3, $3\sigma_v$

C_3 : Axis passes through N and H space.

σ_v : Three vertical plane along three N-H bond containing C_3 axis.

[No horizontal plane]

Point group :

NH_3

Any special group?

NO

C_n ?

Yes, $n = 3$

$3C_2 \perp C_3$?

NO,

σ_h ?

NO

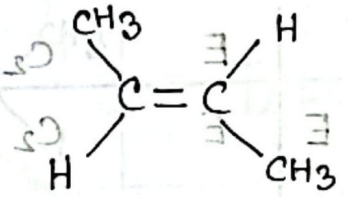
$3\sigma_v$?

Yes

C_{3v}

• Butene-2 (trans)

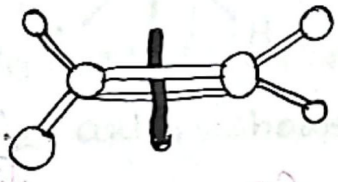
• Configuration:



[Planar]

• Symmetry Elements: E, C_2, σ_h, i

C_2 :

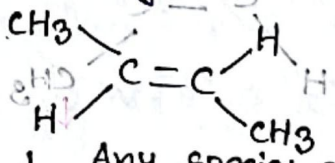


, Axis passes between the bonds.

σ_h : Molecular plane is perpendicular to the C_2 axis. Thus it is horizontal plane.

i : It has opposite groups.

• Point group :



Any special group ?

NO

Cn ?

yes. n=2

$2\,C_2 \perp C_2$ = ?

NO

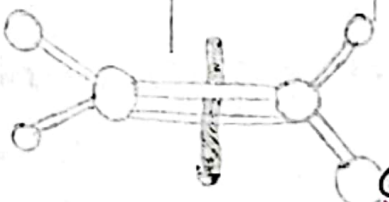
σ_h ?

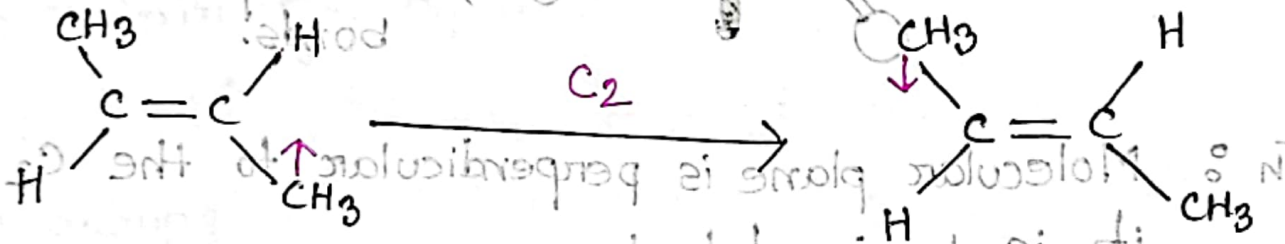
yes

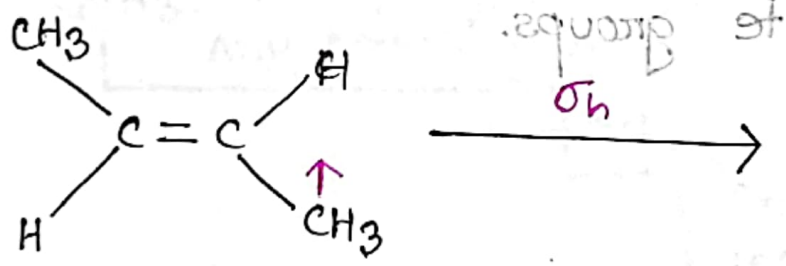
C_{2h}

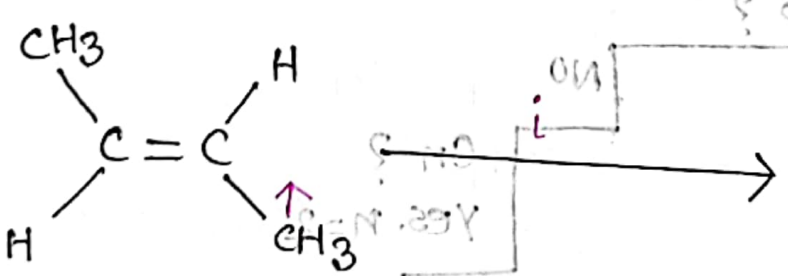
Multiplication table :

	E	C_2	σ_h	i
E	E	C_2	σ_h	i
C_2	C_2	E	i	σ_h
σ_h	σ_h	i	E	C_2
i	i	σ_h	C_2	E



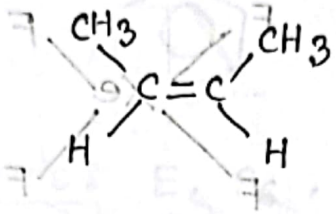






• Butene-2 (cis) :

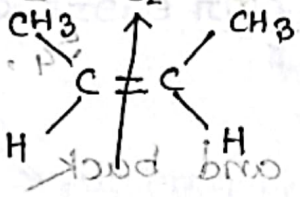
• Configuration :



Planar

• Symmetry Elements : E, C_2, σ, σ'

C_2 :

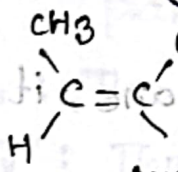


, Axis parallel with paper

σ : Along C_2 axis, shows mirror image : vertical plane

σ' : Molecular plane containing C_2 axis, vertical plane

• Point group : C_{2v}



Any special point group?

NO

C_n?

Yes, $n = 2$

$2\ C_2 \perp C_2$?

NO.

σ_h ?

NO

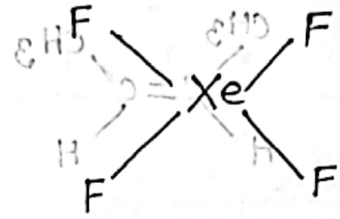
$2\ \sigma_v$?

Yes

C_{2v}

• Xenone tetrafluoride : [XeF_4]

• Configuration:



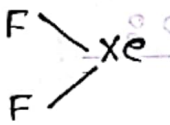
[Planar]

• Symmetry Elements: E, C_4, C_2, $2C_2'$, $2C_2''$, σ_h, $2\sigma_v$, $2\sigma_d$, S_4, i

C_4 : Axis passes forth and back along Xe. [pependicular]

C_2 : Same as C_4.

C_2' : Two axis along two F—Xe—F bonds.

C_2'' : Two axis between two  Bond.

σ_h : As C_4 is pependicular to molecular plane; it is horizontal plane.

σ_v : Two plane containing two C_2' axis ; verctical plane

σ_v' : Two plane containing two C_2'' axis ; verctical plane.

• Point group:

XeF_4 Any special group?

NO

C_n ?
yes, $n=4$

$4\ C_2 \perp C_4$?
yes,

σ_h ?
Yes

D_{4h}

• **Benzene : (C_6H_6)**

• **Configuration :** [planar]

• **Symmetry Elements :** E, S_6, C_6, C_3, C_2, $3C_2'$, $3C_2''$, σ_h, $3\sigma_v$, $3\sigma_v'$, i

C_6 : Middle of benzene ring, perpendicular to paper.

C_3 : Along C_6.

C_2 : Along C_6.

C_2' : Along three opposite corners.

C_2'' : Between three C–C bonds.

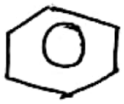
σ_h : C_6 perpendicular to molecular plane ; i.e. horizontal plane.

σ_v : Three planes containing C_2' axis; vertical plane

σ_v' : Three planes containing C_2'' axis; vertical plane.

S_6 : Along C_6 i : Along C_6.

• **Point group :**



Any special group?

No,

C_n ?

yes, $n=6$

$6\,C_2 \perp C_6$?

yes

σ_h ?

yes

D_{6h}

• Cyclopentadienyl anion :

. Configuration : [planar]

. Symmetry Elements : E, C_5, $5C_2$, S_5, σ_h, $5\sigma_v$

C_5 : Axis through neg sign, perpendicular to paper.

C_2 : Along five eg. edges.

S_5 : Along C_5.

σ_h : Molecular plane perpendicular to C_5 ; horizontal plane.

σ_v : Five planes containing five C_2 axis ; vertical plane.

. Point group :

Any special group?

NO

C_n ?

yes, $n = 5$

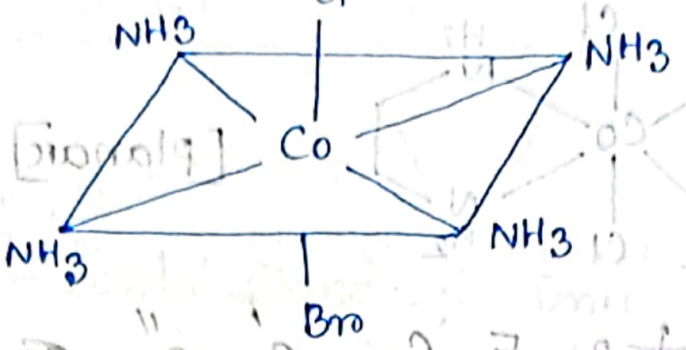
$5C_2 \perp C_5$?

Yes

σ_h ?

yes

D_{5h}



[Planar]

Symmetry Elements: $E, C_4, C_2, 2\sigma_d, 2\sigma'_d$

C_2 : Along perpendicular to paper

C_2' : Along Co and Br.

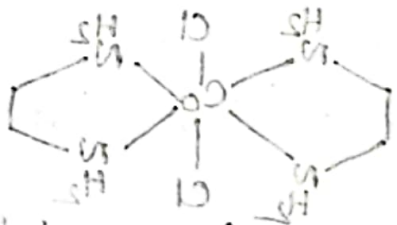
C_2'' : Between two straight lines.

σ_h : C_2 perpendicular to molecular plane; Horizontal

σ : Along C_2 axis, placed at $Cl-Co-Cl$ bond.

σ' : Along bond bridge, parallel to molecular plane.

Point group:



Any special group?

H_0

C_n ?

Yes, $n = 2$

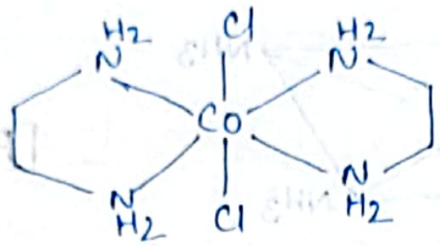
$2C_2$ ⊥ C_2 ?

Yes

D_n ?

Yes

D_{2h}



[planar]

Symmetry Elements: $E, C_2, C_2', C_2'', \sigma_h, \sigma, \sigma'$

C_2 : Along Cl—Co—Cl bond, back and forth to Co, perpendicular to paper

C_2' : Along Co only.

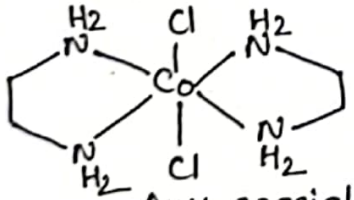
C_2'' : Between two straight lines.

σ_h : C_2 perpendicular to molecular plane; horizontal plane.

σ : Along C_2 axis, placed at, Cl—Co—Cl bond.

σ' : Along Co bond bridge, parallel to molecular plane.

Point group:



Any special group?

NO

C_n ?
Yes, $n = 2$

$2\ C_2 \perp C_2$?
yes

σ_h ?
yes

D_{2h}

POINT GROUP

- $nC_2 \perp C_n$
  - (i) Horizontal plane → Dnh
  - (ii) n verticle plane $\perp C_n$ → Dnd
  - (iii) no plane → Dn
- C_n
  - (i) Horizontal plane → Cnh
  - (ii) n verticle plane $\perp C_n$ → Cnv
  - (iii) no plane → Cn
- Linear
  - (i) Symmetrical → $D_{\infty h}$
  - (ii) Asymmetrical → $C_{\infty v}$
- Special:
  - (i) Octahedral → O_h
  - (ii) Tetrahedral → Td
  - (iii) Icosahedral → Ih
- 2 symmetry element
  - (i) E & i → Ci
  - (ii) E & σ → Cs
  - (iii) E & Sn → Sn
- Only identity → C_1
- staggard → Dnd
- Eclipsed → Dnh

Definition:

- **Symmetry Operation:** A symmetry operation is a movement of a body (molecule) such that after the movement has been carried out, every point of the body is coincident with an equivalent point (or perhaps the same point) of the body in it's original orientation.

- **Symmetry element:** A symmetry element is a geometrical entity such as a line (axis), a plane or a point with respect to which one or more symmetry operation may be carried out.

- **Proper rotation axis: / n-fold rotation axis: (C_n)**
If a rotation of a molecule by an angle of $2\pi/n$ about an axis gives an indistinguishable configuration, then the axis is called an axis of rotation / proper rotation axis / n-fold rotation axis of symmetry.

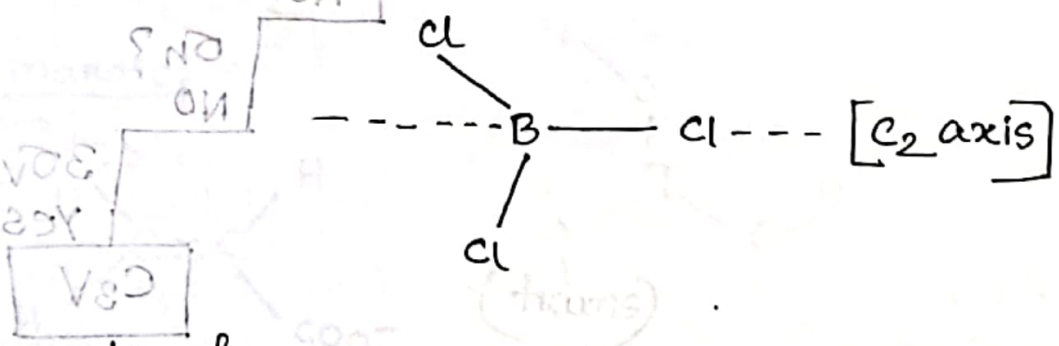
- ~~Molecular sy~~

- **Center of symmetry (i):** A molecule posses a center of symmetry if every atom when moved in a straight line through the center and equal distance on the other side of the center, encounter a similiar atom.

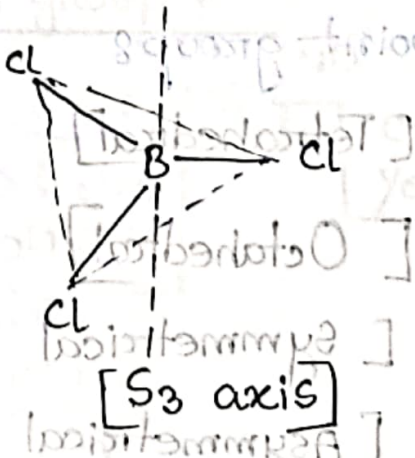
• **Improper rotation axis: (S_n)** If a rotation of a molecule by an angle of $2\pi/n$ about an axis followed by reflection through a plane perpendicular to that axis gives an indistinguishable configuration, then the axis is called an improper axis of rotation / n-fold improper axis of symmetry.

• **Plane of symmetry:** If reflection of all parts of a molecule through a plane gives an indistinguishable orientation, then the plane is called plane of symmetry.

Example of proper rotation axis:

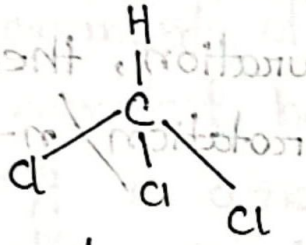


Example of improper rotation axis:



Point group :

- $CHCl_3$



Does it belong to special point group?

NO

$C_n = ?$

yes, $n = 3$

$3C_2 \perp C_2$?

No.

σ_h ?

NO

$3\sigma_V$?

yes

C_{3V}

- Name of special point group :

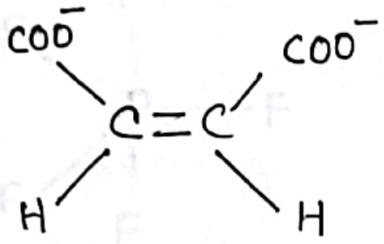
$CBr_4 \rightarrow T_d$ [Tetrahedral]

$SF_6 \rightarrow O_h$ [Octahedral]

$CO_2 \rightarrow D_{\infty h}$ [Symmetrical linear]

$HCl \rightarrow C_{\infty v}$ [Asymmetrical linear]

- Fumarate ion $(O_2CHC=CHCO_2)^{2-}$



(cis)

Does this molecule belong to special point group?

NO

C_n = ?

yes, n = 2

$2C_2 \perp C_2$?

NO

σ_h ?

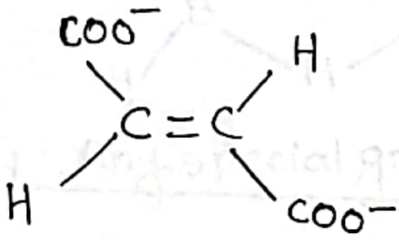
NO

$2\sigma_v$?

yes

C_{2V}

Fumarate:



(trans)

Any special group ?

NO

C_n = ?

yes, n = 2

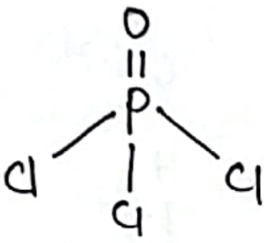
$2C_2 \perp C_2$?

NO

σ_h ?

yes

C_{2h}



Any special group?

NO

$C_n = ?$

yes, $n = 3$

$3C_2 \perp C_3$?

NO

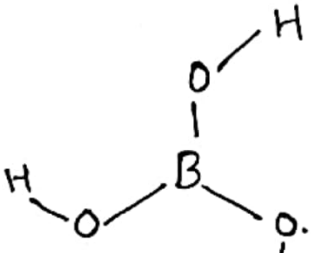
$\sigma_h = ?$

NO

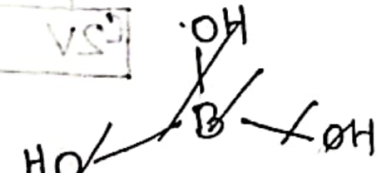
$3\sigma_v$?

yes

C_{3v}



H_3BO_3



Any special group?

NO

$C_n = ?$

yes $n = 3$

$3C_2 \perp C_3$?

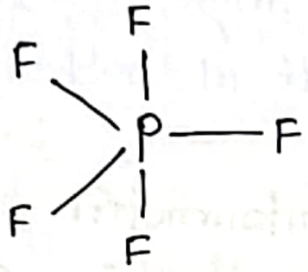
~~yes~~ No

σ_h ?

yes

C_{3h}

- PF_5



Any special group?

NO

C_n=?

yes, n=3

$3C_2 \perp C_3$?

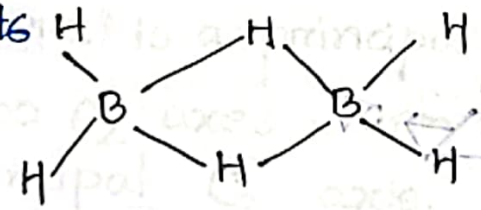
yes.

σ_h ?

yes

D_{3h}

- B_2H_6



Any special group?

NO

C_n=?

yes, n=2

$2C_2 \perp C_2$?

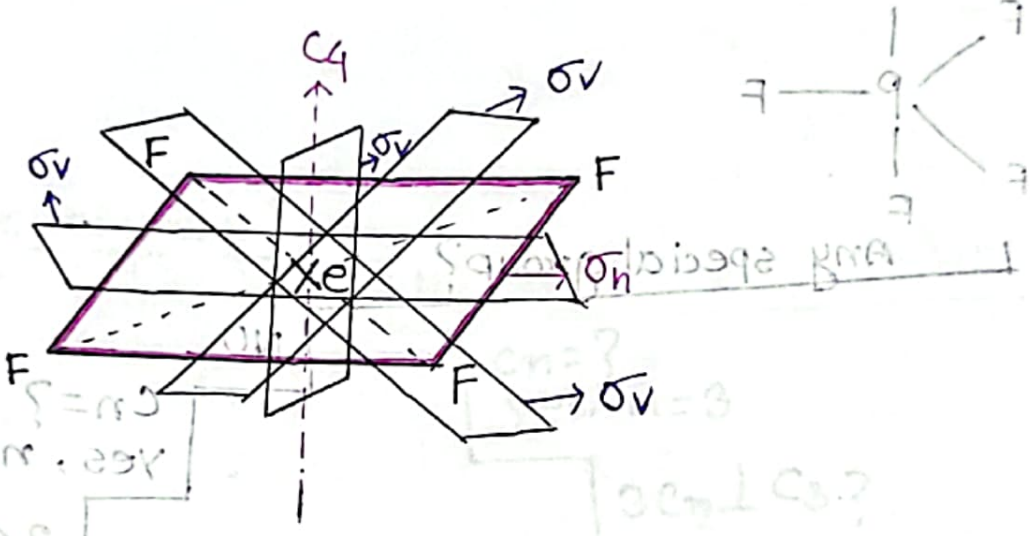
yes

σ_h ?

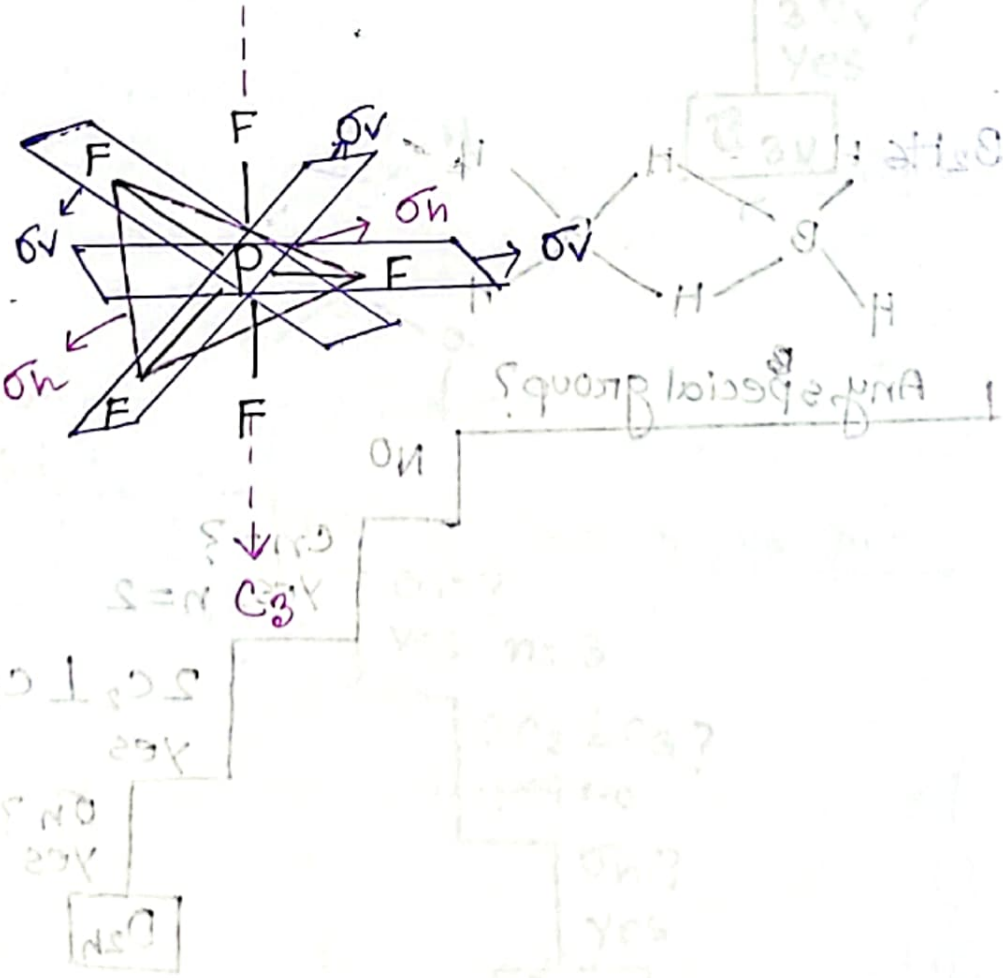
yes

D_{2h}

- Show the C_4 axis, σ_h and σ_v in XeF_4.



- Show the C_3 axis, σ_h and σ_v in PF_5.



- Suppose a molecule belongs to the point group D_{4h}/C_{2v}. What information about the molecular symmetry are embedded in the point group symbol D_{4h}/C_{2v}?

D_{4h}:

The informations embedded in the point group symbol D_{4h} a

a) There ~~are~~ is a principal rotation axis of order four.

b) Four C_2 axes are in a perpendicular position with C_4 axis.

c) h represents the molecule having a horizontal plane

C_{2v}:

a) There is a principal rotation axis of order two.

b) Two C_2 axes doesn't exist at right angles to the principal C_2 axis.

c) v represents two vertical planes containing principal rotation axis.

- What is the difference between vertical plane and horizontal plane?

- The plane containing the principal rotation axis is called vertical plane; whereas the plane at the right angle to the principal rotation axis is called a horizontal plane.

• What is the condition that needs to be fulfilled if the two symmetry operations belong to the same class?

Two operations A and B are in the same class if there is some operation X such that:

$$XAX^{-1} = B$$

[X^{-1} is the inverse of X; i.e. $XX^{-1} = E$]

We say that B is the similarity transform of A, and that A and B are conjugate.

• Write the 'Reduction Formula' that is used to reduce the reducible representation to their component irreducible representations.

The 'Reduction Formula' is:

Number of times an irreducible representation occurs in the reducible representation $= \frac{1}{h} \sum_{\text{over all classes}} \chi_R \times \chi_I \times N$

where, h = order of the group (number of operations in the group)

χ_R = character of the reducible representation.

χ_I = character of the Irreducible representation.

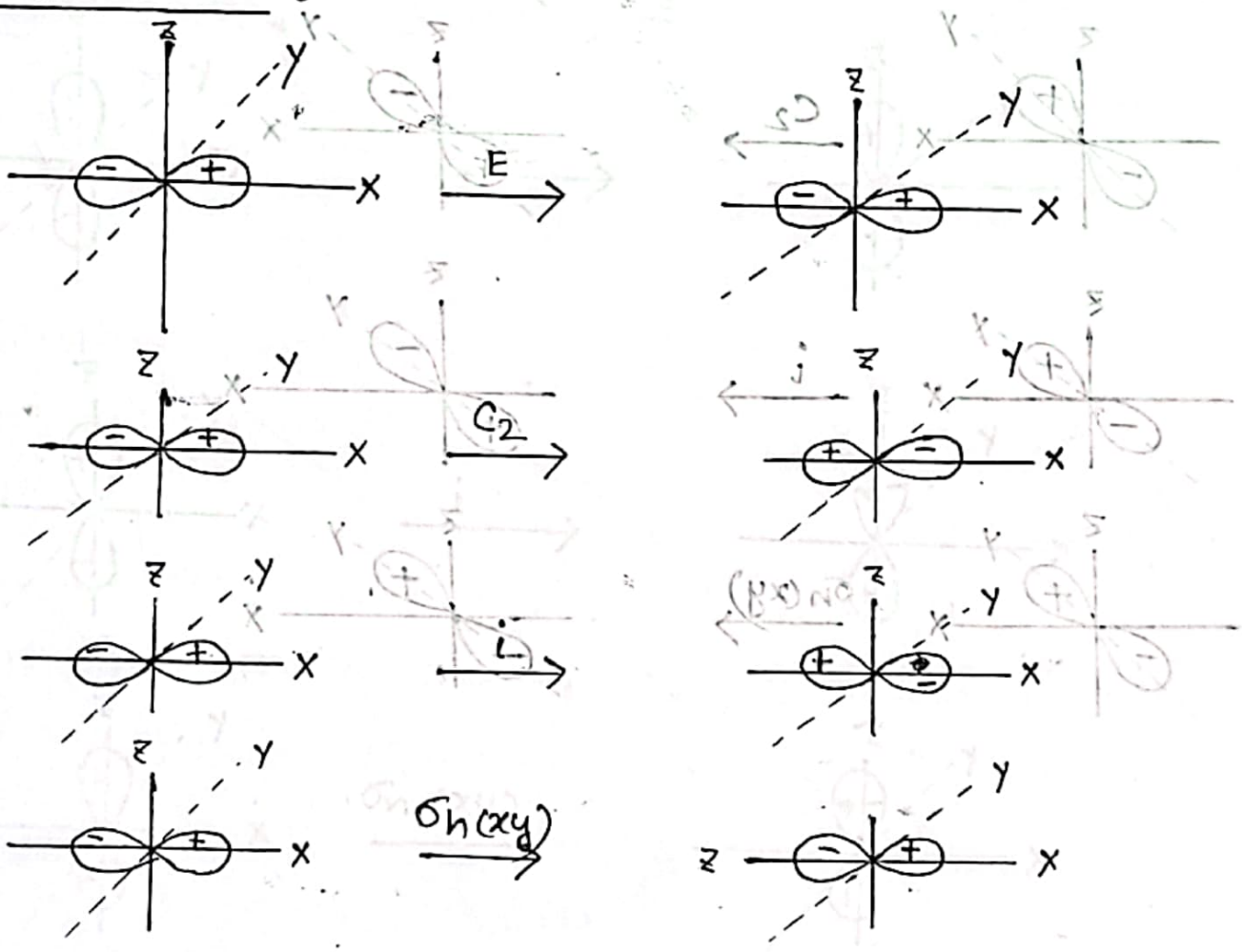
N = Number of symmetry operations in the class

Taking the C_2 axis as the z axis and σ_h to be the xy plane, to what representations do the p_x, p_y, p_z / d_{xy}, d_{xz} and d_{yz} orbitals belong in C_{2h} symmetry. The character table of C_{2h} point group is.

C_{2h}	E	C_2	i	$\sigma_h(xy)$
Ag	1	1	1	1
Bg	1	-1	1	-1
Au	1	1	-1	-1
Bu	1	-1	-1	1

Ans:

Px orbital:



Px orbital,

E is represented by 1

C_2 " " " 1

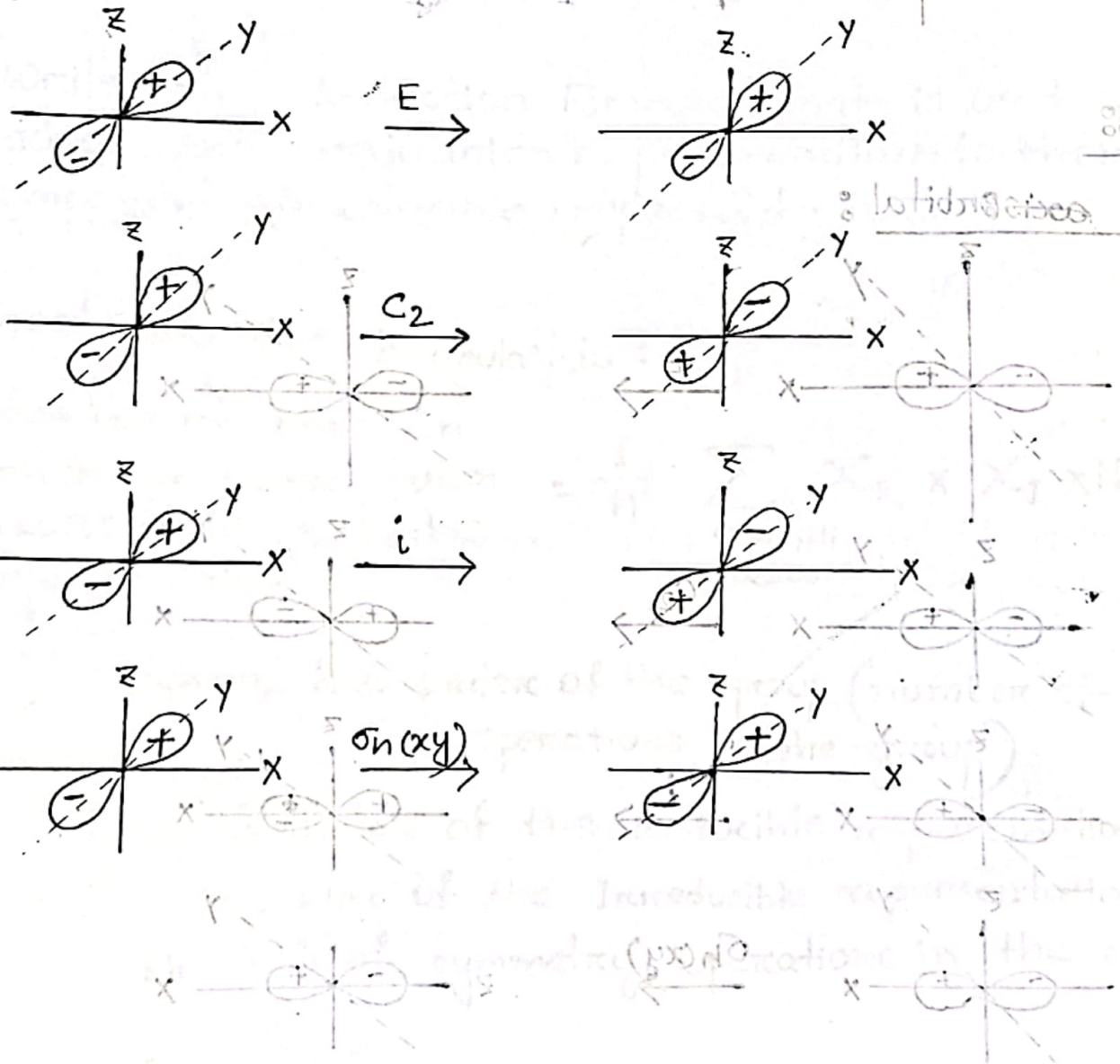
i " " " −1

σ_h " " " 1

Thus Px orbital belongs to B_u representation.

C_{2h}	E	C_2	i	$\sigma_h(xy)$
B_u	1	−1	−1	1

Py orbital :



E is represented by 1

C_2 " " " -1

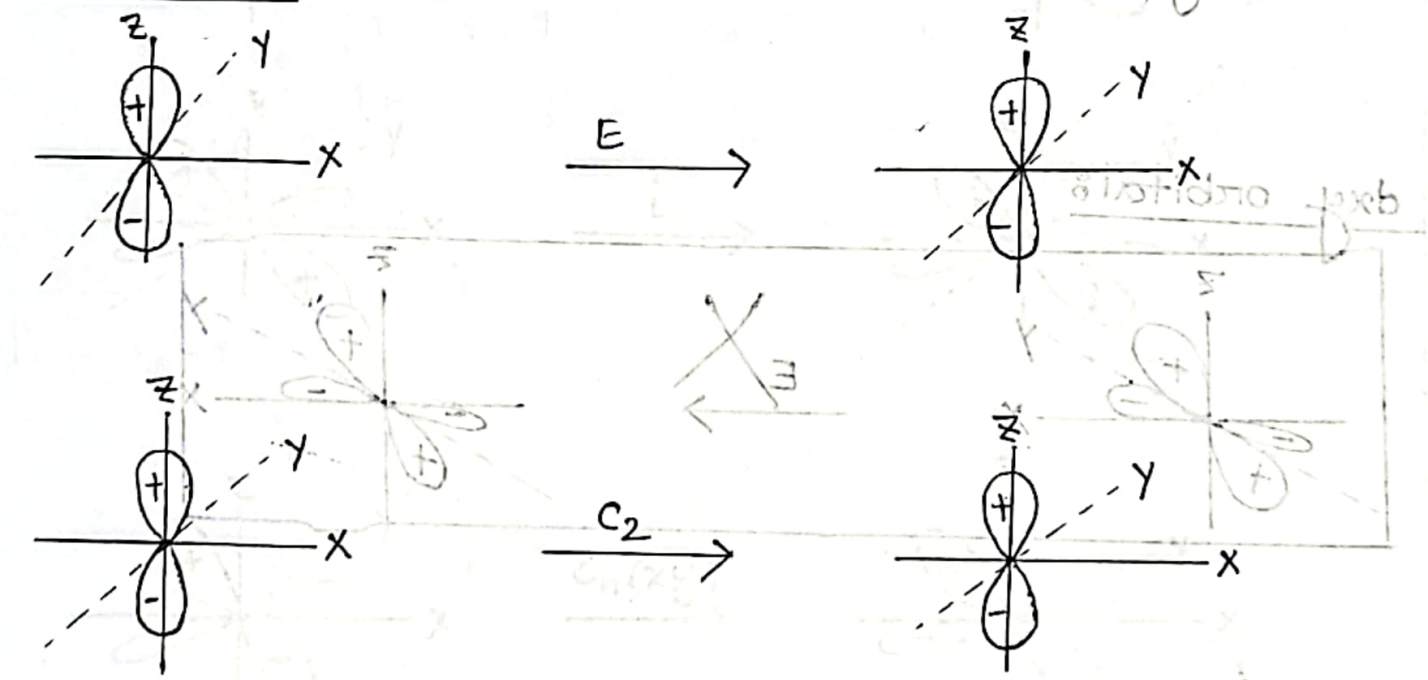
i " " " -1

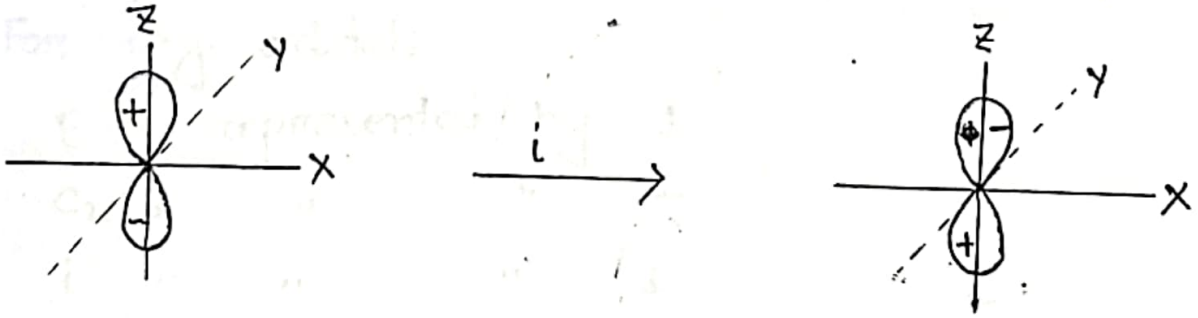
σ_h " " " 1

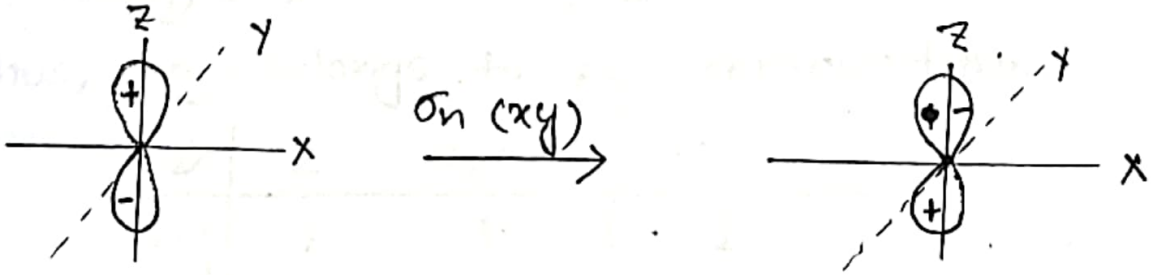
Thus P_y orbital belongs to Bu representation.

C_{2h}	E	C_2	i	σ_h (xy)
Bu	1	-1	-1	1

P_z orbital:







For p_z orbital,

E is represented by 1

C_2 " " " 1

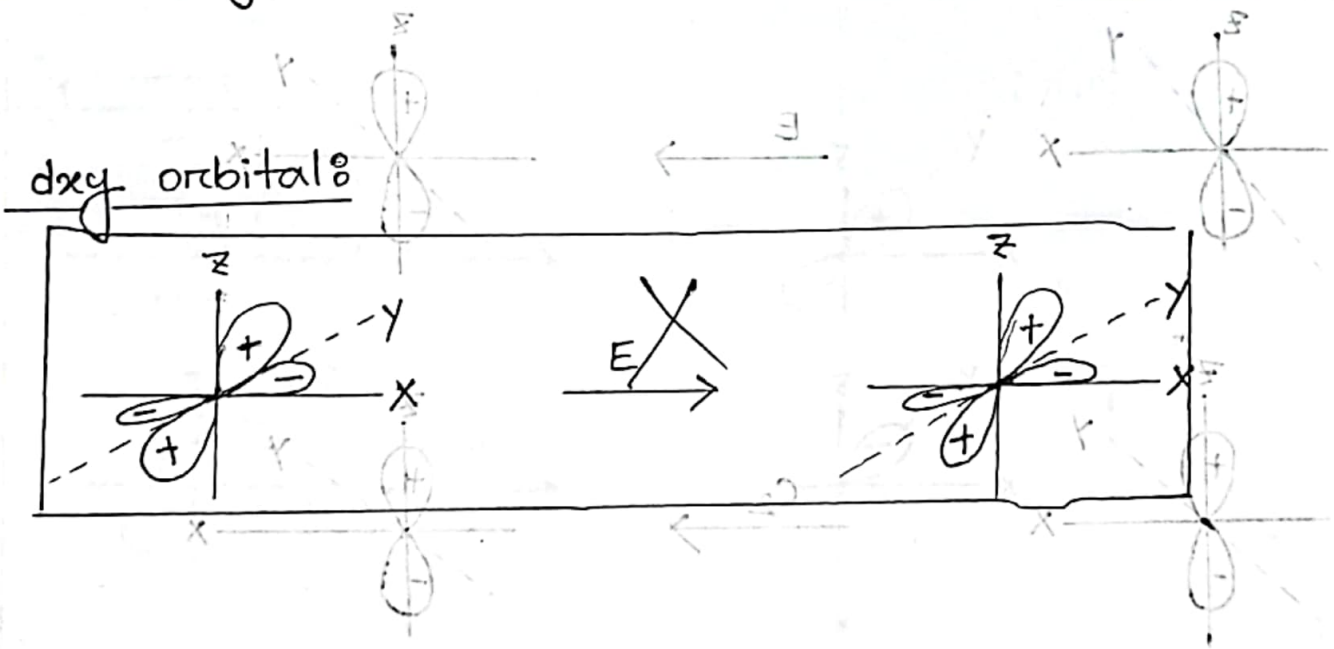
i " " " −1

σ_h " " " −1

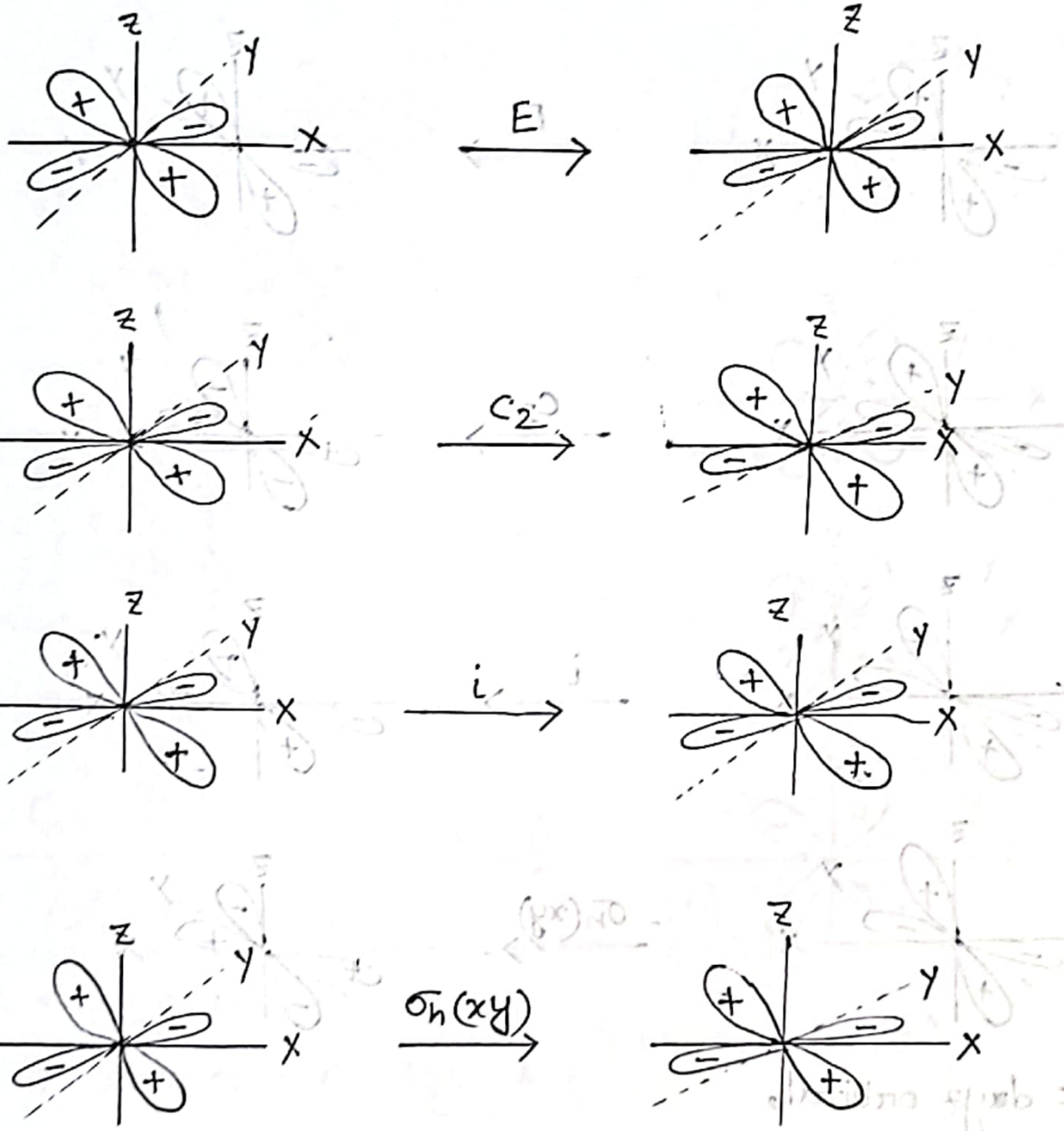
Thus P_z orbital belongs to A_u representation.

C_{2h}	E	C_2	i	$\sigma_h(xy)$
A_g	1	1	−1	−1

dxy orbital:



dxy orbitals



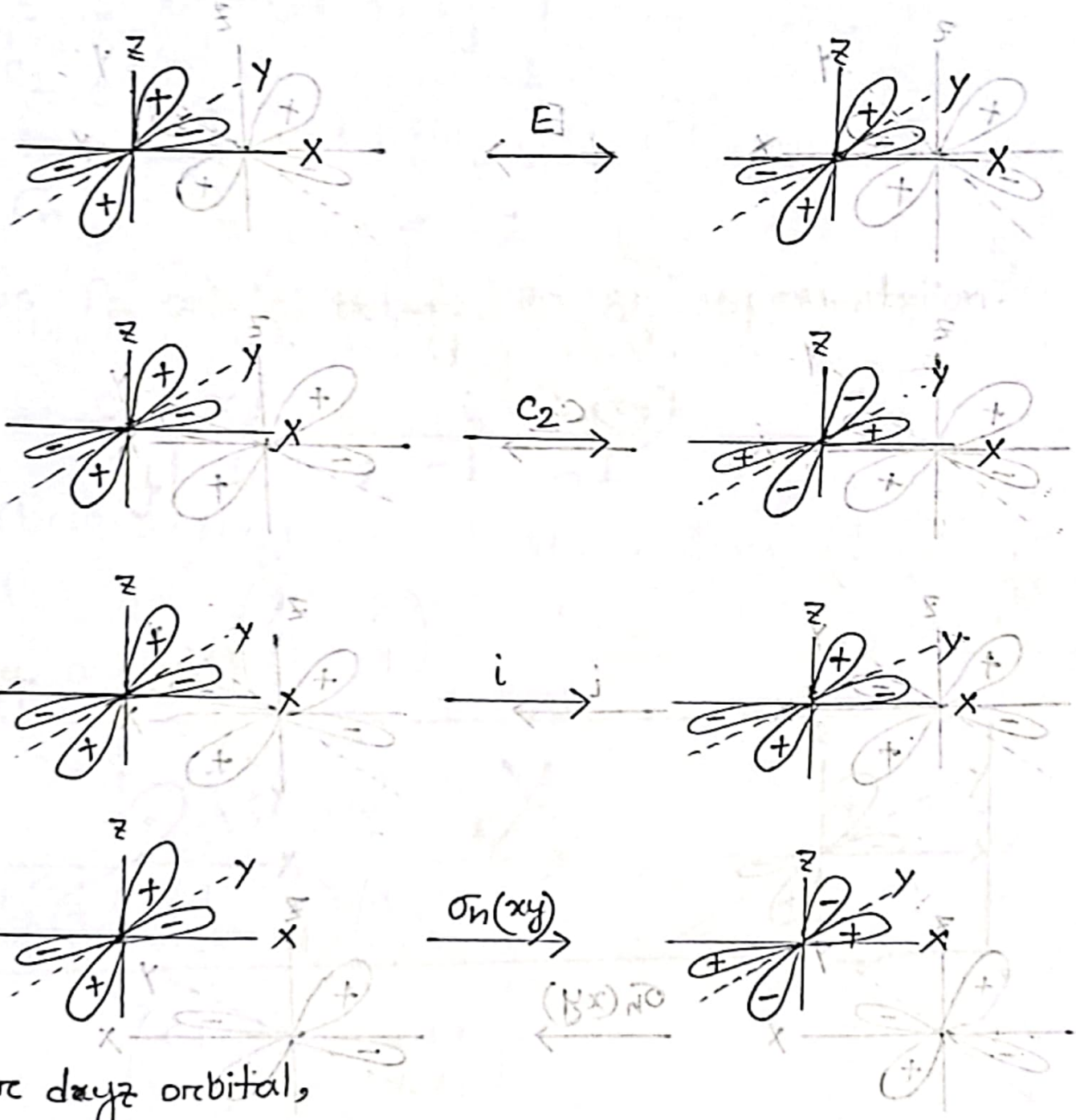
For dxy orbital,

E is represented by 1
C_2 " " " 1
i " " " 1
$\sigma_h(xy)$ " " " 1

Thus, dxy belongs to Ag representation.

C_{2h}	E	C_2	i	σ_h
Ag	1	1	1	1

d_{yz} orbital:



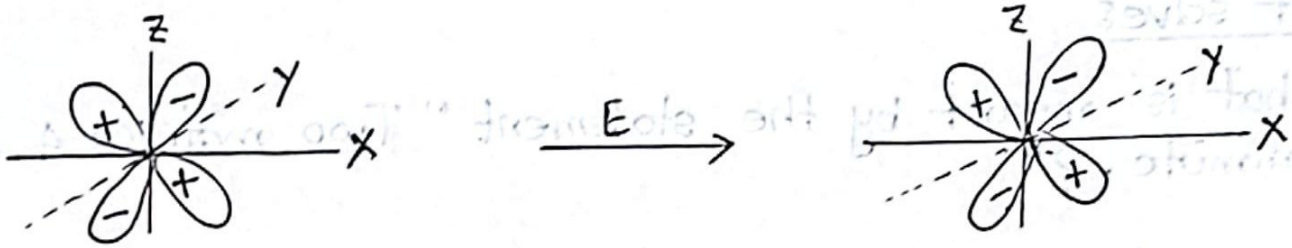
For d_{yz} orbital,

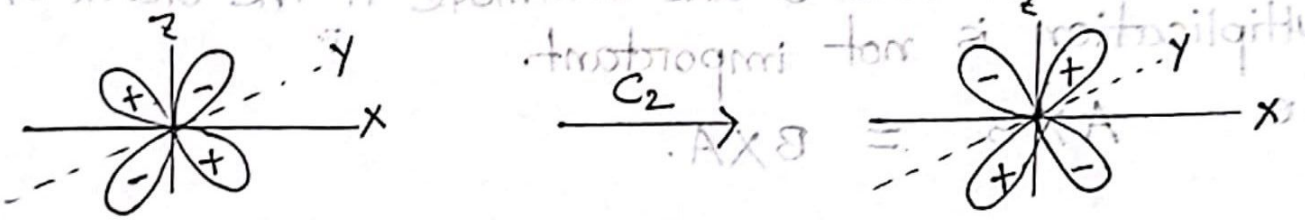
E is represented by 1
C_2 " " " -1
i " " " 1
σ_h " " " -1

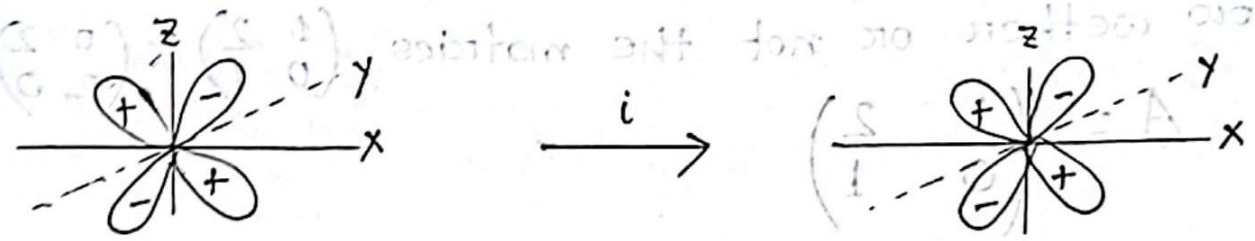
Thus, d_{yz} orbital belongs to B_g point group.

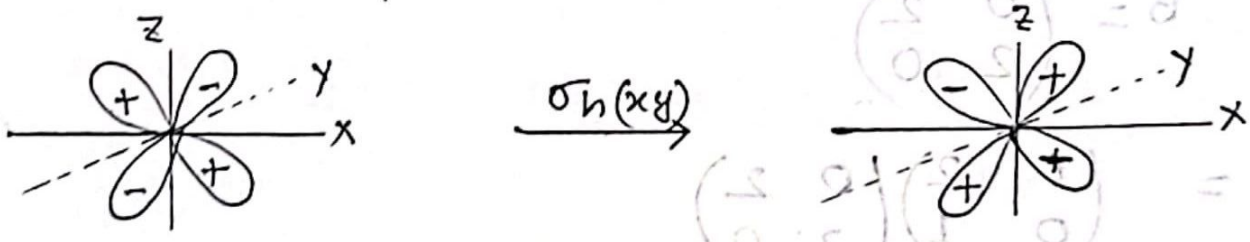
C_{2h}	E	C_2	i	σ_h
B_g	1	-1	1	-1

d_{xz} orbital :









For d_{xz} orbital,

E is represented by 1

C_2 " " " -1

i " " " 1

σ_h " " " -1

Thus, d_{xz} orbital belongs to B_g representation.

C_{2h}	E	C_2	i	σ_h
B_g	1	−1	1	−1

Test solve:

- What is meant by the statement "Two matrices A and B commute. ?"

Two matrices A and B ~~are~~ commute if the order of multiplication is not important.

i.e: $AXB \equiv BXA$.

- Show wether or not the matrices $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ & $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$ a

Let, $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

$B = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$

$$AXB = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0+4 & 2+0 \\ 0+2 & 0+0 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 2 \\ 2 & 0 \end{pmatrix}$$

$$BXA = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0+0 & 0+2 \\ 2+0 & 4+0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2 \\ 2 & 4 \end{pmatrix}$$

As, $AXB \neq BXA$, They do not commute

CHAPTER 4 - MATRICS

Test solve:

- What is meant by the statement "Two matrices A and B commute. ?"

Two matrices A and B ~~are~~ commute if the order of multiplication is not important.

i.e: $AXB \equiv BXA$.

- Show wether or not the matrices $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ & $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$ commute.

Let, $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

$B = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$

$$AXB = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0+4 & 2+0 \\ 0+2 & 0+0 \end{pmatrix}$$

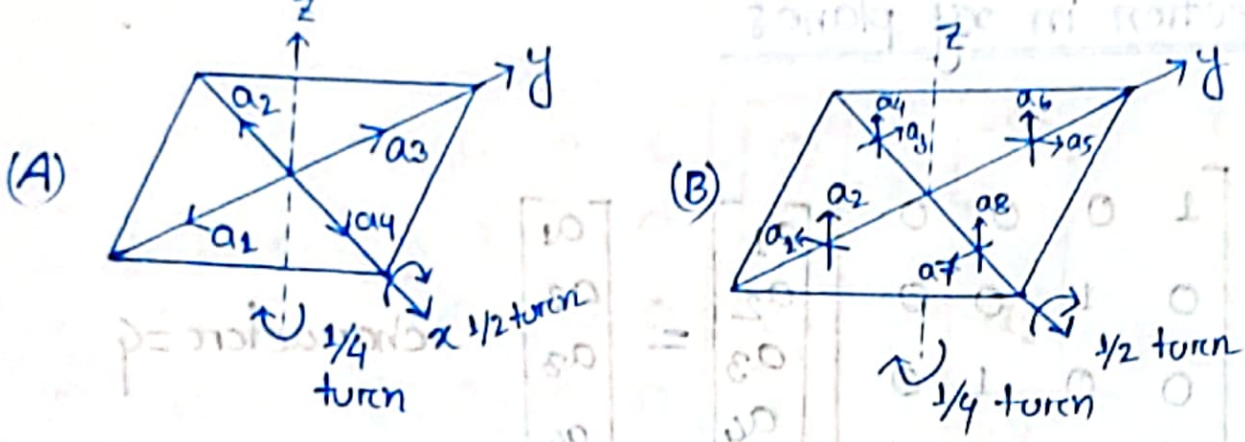
$$= \begin{pmatrix} 4 & 2 \\ 2 & 0 \end{pmatrix}$$

$$BXA = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0+0 & 0+2 \\ 2+0 & 4+0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2 \\ 2 & 4 \end{pmatrix}$$

As, $AXB \neq BXA$, They do not commute



1/4 th turns

(A)

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} a_4 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

character = 0

$$\begin{bmatrix} & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \end{bmatrix} = \begin{bmatrix} & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \\ a_8 \end{bmatrix} = \begin{bmatrix} & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \end{bmatrix}$$

• Reflection in xy plane:

(A)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

character = 4

(B)

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \\ a_8 \end{bmatrix} = \begin{bmatrix} a_1 \\ -a_2 \\ a_3 \\ -a_4 \\ a_5 \\ -a_6 \\ a_7 \\ -a_8 \end{bmatrix}$$

character = 0

• 1/2 turn: (About x axis)

(A)

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} a_3 \\ a_2 \\ a_1 \\ a_4 \end{bmatrix}$$

character = 2

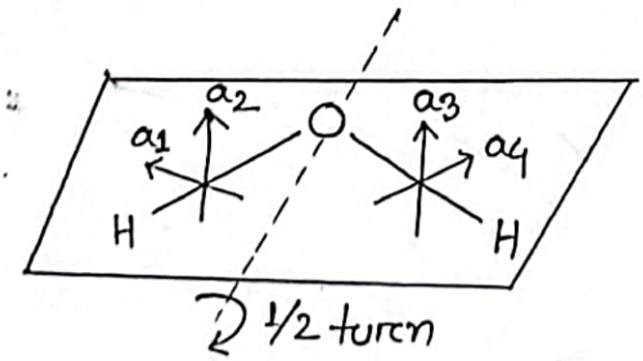
(b)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \\ a_8 \end{bmatrix} = \begin{bmatrix} -a_5 \\ -a_6 \\ -a_3 \\ -a_4 \\ -a_1 \\ -a_2 \\ -a_7 \\ -a_8 \end{bmatrix}$$

character $= -4$

Generate a matrix representation of two-fold rotation for H_2O molecule, using the four arrows and a matrix representation of reflection in the plane of the molecule. Also find their characters.

Answer:



Two fold rotation:

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} a_4 \\ -a_3 \\ -a_2 \\ a_1 \end{bmatrix}$$

character $= 0$

Reflection in planes

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} a_1 \\ -a_2 \\ -a_3 \\ a_4 \end{bmatrix}$$

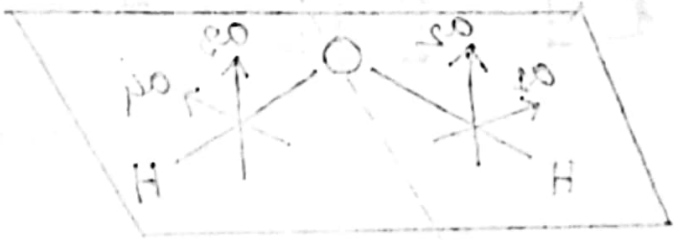
character = 0

Set up the Matrices which will peform the following transformation:

(A) $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} -y \\ -x \end{pmatrix}$ & (B) $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to $\begin{pmatrix} -y \\ x \\ -z \end{pmatrix}$

Answer:

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ -x \end{pmatrix} \quad \& \quad \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -y \\ x \\ -z \end{pmatrix}$$

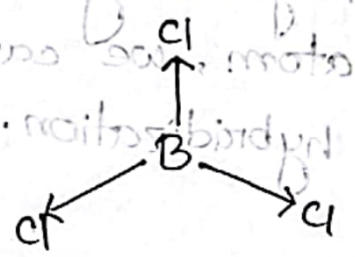


• Finding out the hybrid orbitals

Find the hybrid orbitals of a central atom suitable forming a set of BCl_3 bonds. [or BF_3 or BH_3]

Answer:

Structure:



Let us consider, the three arrows representing th
of BCl_3 molecule.
The point group of BCl_3 is D_{3h}. The complete set of c
of the reducible representation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ:	3	0	1	3	0	1

character table of

According to the point group D_{3h}:

No of $A_1' = \frac{1}{12}(3+3+3+3) = 1$

No of $A_2' = 1/12\,(3-3+3-3) = 0$

No of $E' = 1/12\,(6+0+6+0) = 1$

No of $A_1'' = 1/12\,(3+3-3-3) = 0$

No of $A_2'' = 1/12\,(3-3-3+3) = 0$

No of $E'' = 1/12\,(6+0-6+0) = 0$

The irreducible representation is, $\Gamma = A_1'' + E'$

For hybrid orbitals,

S or d_{z^2} belongs to symmetry species A_1'.

(p_x, p_y) or $(d_{x^2-y^2}, d_{xy})$ belongs to symmetry species E'.

Since, there is no energetically favourable d orbital in the B (boron) atom, we can't neglect $d_{x^2-y^2}$, d_{xy} and d_{z^2} orbitals for hybridization.

So, suitable orbitals are:

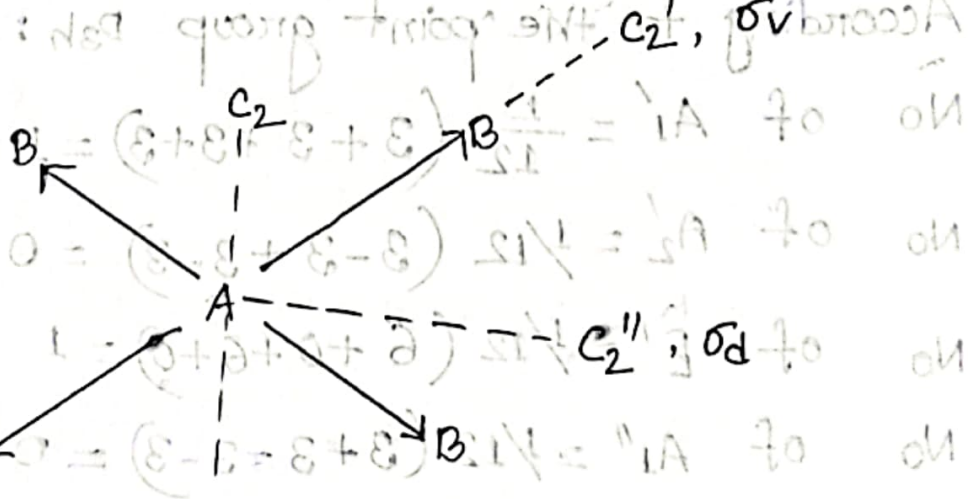
$A_1' \longrightarrow s$

$E_1' \longrightarrow p_x, p_y$

Thus, the most likely set of hybrid orbitals are sp^2.

Find the hybrid orbitals of a central atom suitable for forming a set of square planar bonds. Use the D_{4h} character table.

Answers



Let us consider, the three arrows representing the basis of the square planar molecules.

The point group of square planar molecule is D_{4h}. The complete set of characters of the reducible representation is,

D_{4h}	E	$2C_4$	$(C_2=C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	4	0	0	2	0	0	0	4	2	0

According to the character table of D_{4h} point grou

No of, [shortcut]

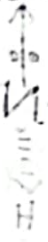
$A_{1g} = \frac{1}{16}(4+4+4+4) = 1$

$A_{2g} = 0$

$B_{1g} = 1$

$B_{2g} = 0$

$E_g = 0$



$A_{1u} = 0$

$B_{2u} = 0$

$B_{1u} = 0$

$B_{2u} = 0$

$E_u = 1$

The irreducible representation is, $\Gamma = A_{1g} + B_{1g} + E_u$

For hybrid orbitals we can see from character to

s or dz^2 belongs to species A_{1g};

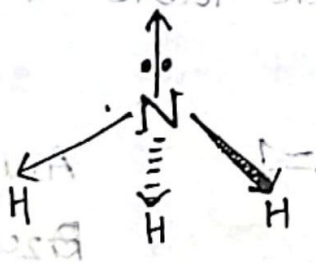
dx^2-y^2 " " " B_{1g}

Px and Py " " " E_u.

Hence, the set of hybrid orbitals are, dsp^2

Find the hybrid orbitals of a central atom suitable for forming a set of trigonal pyramidal bond For NH_3.

Answer:



Let us consider, the 3 N-H arrows and one lone pair representing the basis of the trigonal pyramidal molecule NH_3.

The point group of NH_3 molecule is C_{3v}. The complete set of characters of the reducible representation is,

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ	4	1	2

According to the characters table of C_{3v} point group,

No of, $A_1 = 1/6\ (4+2+6) = 2$

No of $A_2 = 1/6\ (4+2-6) = 0$

No of $E = 1/6\ (8-2+0) = 1$

The irreducible representation is $= 2A_1 + E$

From characters table we can see that,

s, p_z or d_{z^2} orbital belongs to A_1 representation

(P_x, P_y) or $(d_{x^2-y^2}, d_{xy})$, (d_{xz}, d_{yz}) orbital belongs to E.

Since, there is no energetically favourabled orbital in N atom, we can neglect. d_{z^2}, $d_{x^2-y^2}$, d_{xy}, d_{xz} and d_{yz} orbitals of hybridization.

Suitable orbitals are,

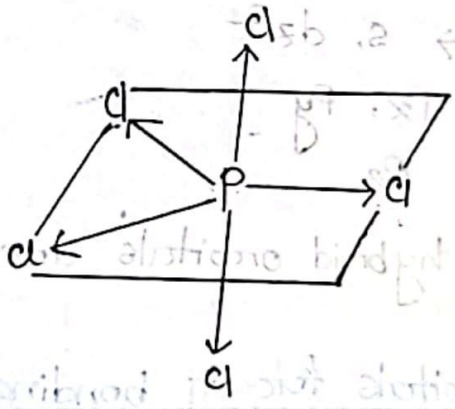
$2A_1 \longrightarrow s, p_z$

$E \longrightarrow p_x, p_y$

Hence, the set of hybrid orbitals are, sp^3

Find out the orbitals of phosphorous (P) atom that will be suitable for forming the five P-F/P-Cl bonds in PF_5 / PCl_5

Ans:



Let us consider, the five arrows representing the basis of PCl_5 molecule.

The point group of PCl_5 molecule is D_{3h}. The complete set of characters of the reducible representation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	5	2	1	3	0	3

According to the characters table of D_{3h} point group,

No of;

$A_1' = 2$

$A_2' = 0$

$E' = 1$

$A_1'' = 0$

$A_2'' = 1$

$E'' = 0$

The reducible irreducible representation is, $\Gamma = 2A_1' + A_2'' + E'$.

S and dz^2 orbital belongs to A_1' representation.

(Px, Py) or (dx^2-y^2, dxy) " " E' "

Pz " " A_2'' "

Suitable orbitals are,

$2A_1' \rightarrow$ S, dz^2

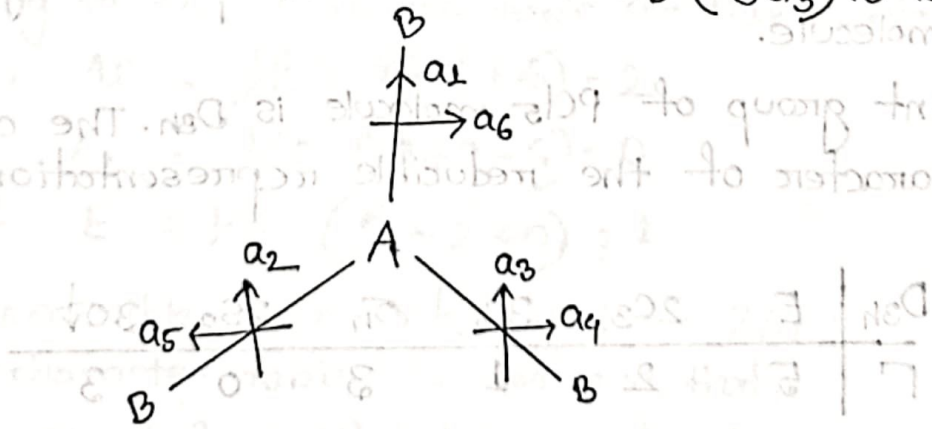
$E' \rightarrow Px, Py$

$A_2'' \rightarrow Pz$

Hence the set of hybrid orbitals are sp^3d.

• Finding out orbitals for π bonding:

Find the orbitals suitable for AB_3 (BCl_3) to form π bonding.



Let us consider the six arrows representing the six p orbitals as a basis of the AB_3 molecule. The arrows: a_1, a_2, a_3 are out of plane and a_4, a_5, a_6 are in the plane.

The point group of AB_3 is D_{3h}. The complete character of the reducible representation is,

For the out of plane set,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ_1	3	0	−1	−3	0	1

For the in plane set,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ_2	3	0	−1	3	0	−1

According to the character table of D_{3h} point g

For out of plane:

No of $A_1' = 0$

$A_2' = 0$

$E' = 0$

$A_1'' = 0$

$A_2'' = 1$

$E'' = 1$

For in plane:

$A_1' = 0$

$A_2' = 1$

$E' = 1$

$A_1'' = 0$

$A_2'' = 0$

$E'' = 0$

So, irreducible representation for out of plane,

$\Gamma_1 = A_2'' + E''$

So, irreducible repres for in plane is,

$\Gamma_2 = A_2' + E'$

From the character table,

Out of plane:

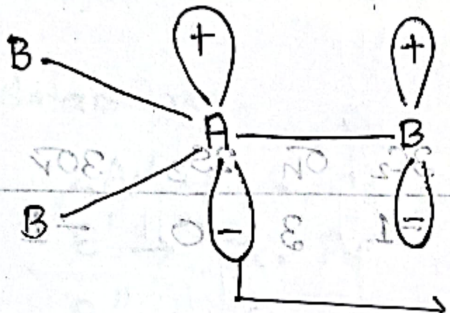
P_z belongs to A_2'' representation

(dxz, dyz) orbital belongs to E'' representation

In plane:

(P_x, P_y) or $(d_{x^2-y^2}, d_{xy})$ orbitals belong to E' representation.

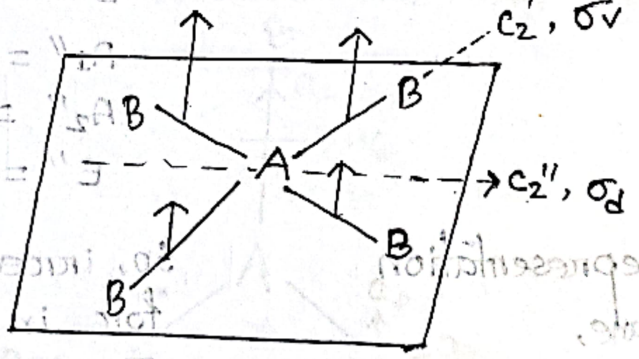
Since, A (Boron) has no energetically available d orbital, we can neglect d_{xz}, $d_{x^2-y^2}$, d_{xy}, d_{yz} for the π bonding. Since, P_x and P_y orbitals are already used for hybridization. only P_z orbital is available for π bonding.



→ unhybridized P_z orbital.

Find the orbitals suitable for the out of plane π-bonding in a square planare molecule.

Answers:



Let us consider the 4 arrows representing the 4-orbitals as a basis of the AB_4 molecule (out of plane).

The point group of the square planare molecule is D_{4h}.

The complete set of the characters of the reducible representation is,

D_{4h}	E	$2C_4$	$C_2(=C_4^2)$	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$
Γ	4	0	0	−2	0	0	0	−4	2	0

According to the character table of D_{4h} point group:

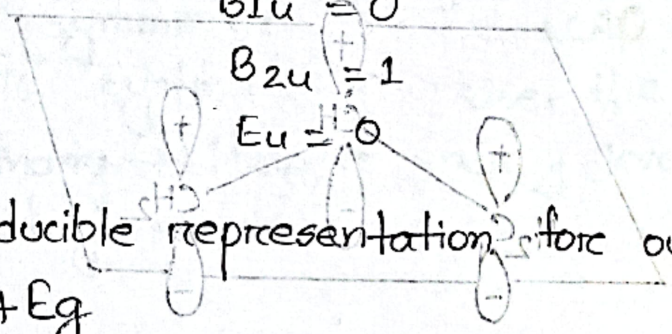
$A_{1g} = 0$ $A_{1u} = 0$

$A_{2g} = 0$ $A_{2u} = 1$

$B_{1g} = 0$ $B_{1u} = 0$

$B_{2g} = 0$ $B_{2u} = 1$

$E_g = 1$ $E_u = 0$



So, the irreducible representation for out of the plane is,

$= A_{2u} + B_{2u} + E_g$

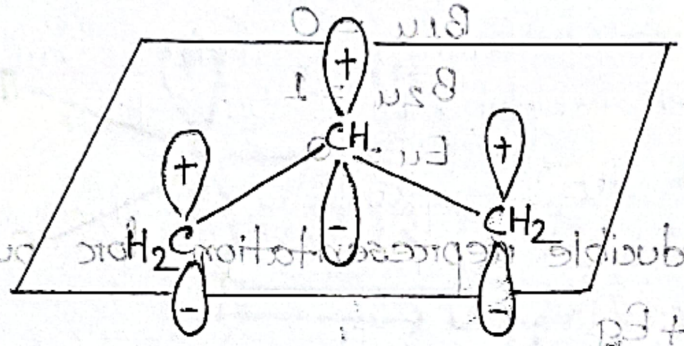
From the character table it is seen that (d_{xy}, d_{yz}) belongs to the E_g representation. p_z belongs to the A_{2u} representation. There is no orbitals in the B_{2u} representation.

Therefore, d_{xy}, d_{yz} and p_z orbitals are suitable for the out of plane π bonding in a square planar molecule.

• LCAO π-molecular orbitals :

How many symmetries of the LCAO π-molecular orbitals of the open chain C_3 system ? Use the C_{2v} character table. How many different energy levels will there be in the symmetry system ?

Answer :



Let us consider the 3-p orbitals representing the basis of this molecule.

The point group of this molecule is C_{2v}. The complete set of all characters of the irreducible representation is,

C_{2v}	E	C_2	σ	σ'
Γ	3	-1	-3	1

According to the point group C_{2v} character table,

No of,

$A_1 = 0$

$A_2 = 1$

$B_1 = 0$

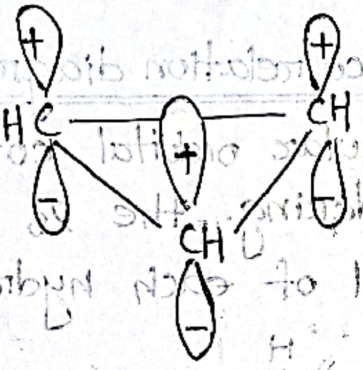
$B_2 = 2$

So, the irreducible representation is $= A_2 + 2B_2$

d_{xy} orbital belongs to A_2 representation.
p_y and d_{yz} " " " B_2 "
None of these orbitals are degenerate. Thus three orbitals with all different energy will be there in the symmetry system.

How many symmetricies of the LCAO-π-molecular orbitals of this system (Fig). Use the D_{3h} character table. How many different energy levels will there be in the symmetry system?

Answers:



cyclopropenyl radical.

Let us consider, three p orbitals representing the basis of the cyclopropenyl radical.

The point group of this molecule is D_{3h}. The complete set of the character of the reducible representation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	3	0	−1	−3	0	1

According to the character table of D3h point group,

[calculation]

The irreducible representation is = $A_2'' + E''$

From the character table,

P_z belongs to A_2'' representation.

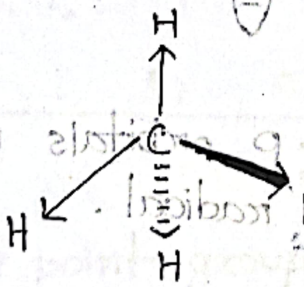
(xz, yz) " " E'' "

These three orbitals consist of a doubly degenerate pair (E'') and one singly degenerate pair orbital (A''). This result tells us nothing about the energy diff. between A_2'' and E'' orbitals.

• Molecular orbital correlation diagram:

Set up the molecular orbital correlation diagram for CH_4 molecule, considering the 2s and 2p orbitals of carbon and 1s orbital of each hydrogen.

Answer:



Let us consider, the four C-H bond as a basis of the CH_4 molecule.

CH_4 molecule belongs to the Td point group. The complete set of character of the reducible representation of the CH_4 molecule is,

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$
Γ	4	1	0	0	2

From the character table of Td point group,

No of,

$A_1 = 1/24\,(4\times1\times1 + 8\times1\times1 + 3\times0\times1 + 6\times0\times1 + 6\times2\times1)$

$= 1$

$A_2 = 0$

$E = 0$

$T_1 = 0$

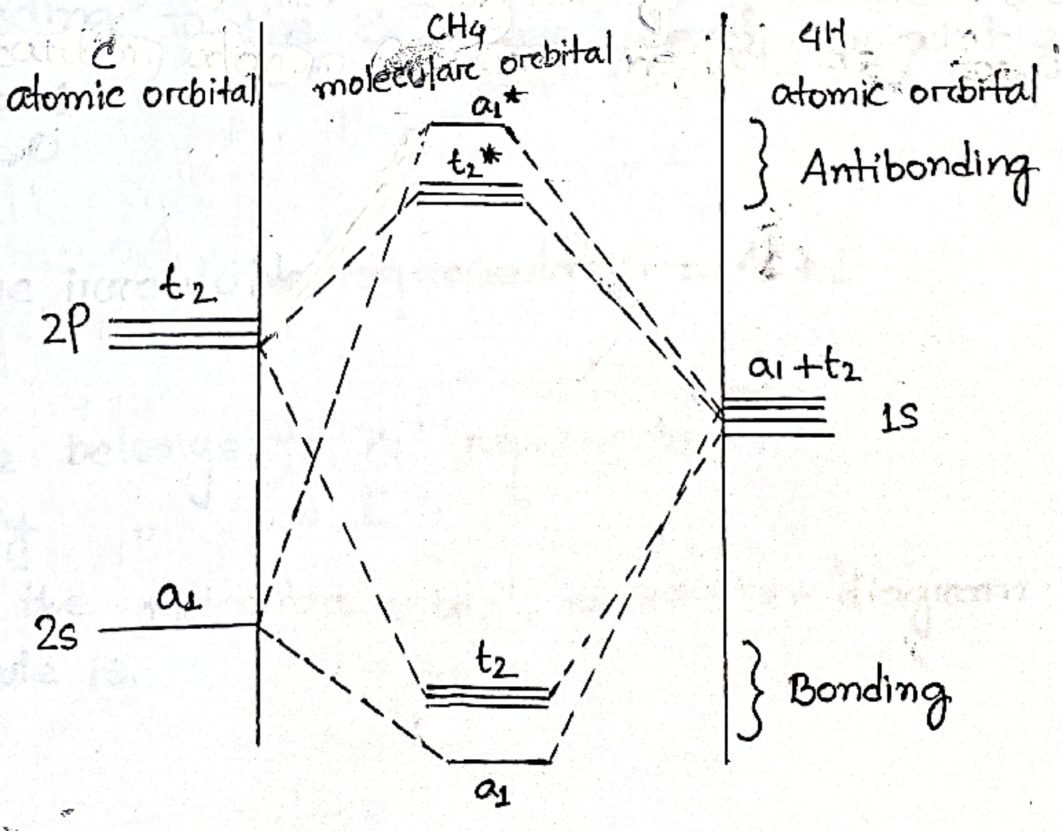
$T_2 = 1$

So, the irreducible representation is $= A_1 + T_2$

S orc dz^2 orbitals belong to A_1

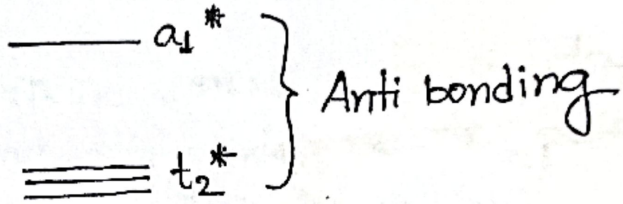
P_x, P_y, P_z " "

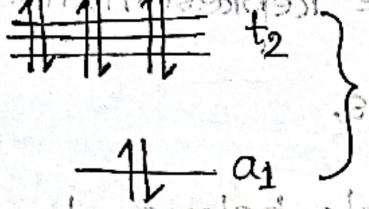
So, the co-rrelation diagrcam of CH_4 molecule is



Here, the total no of electrons is eight; Four electrons from the outermost orbitals of carbon atom and the four electrons from the 4H atoms.

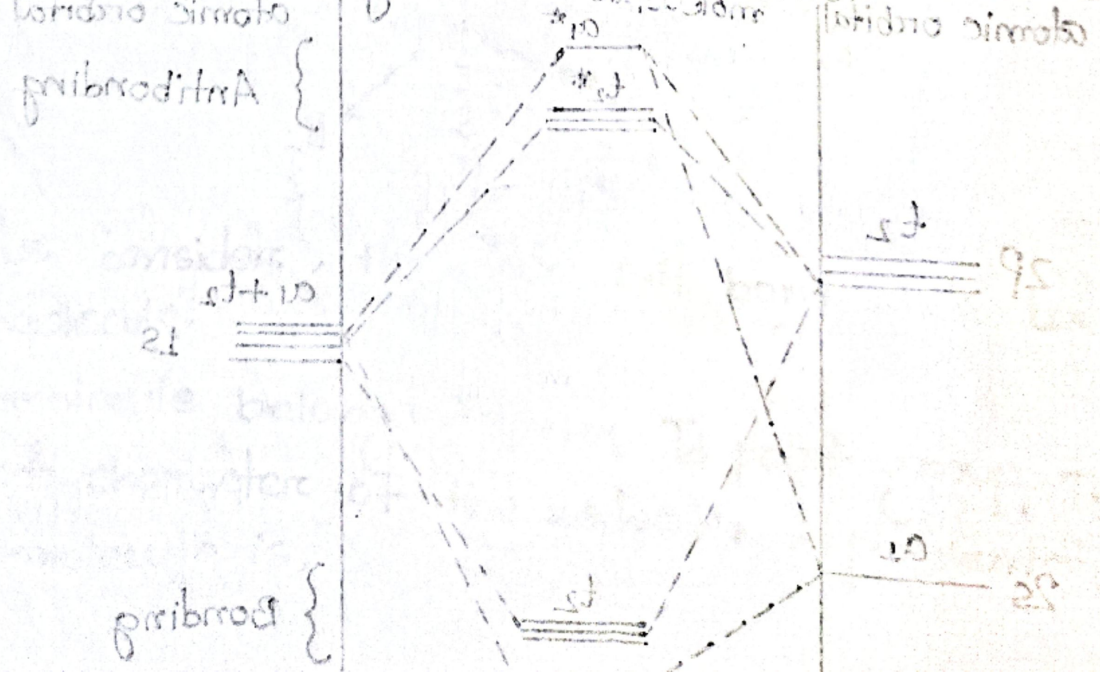
Therefore the electrons are distributed by following manner:





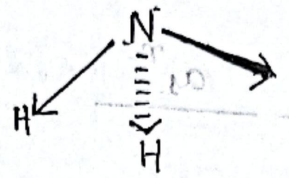
Bond order $= \frac{1}{2}(8-0) = 4$

Bond order is four.



Set up the molecular orbital correlation diagram for the NH_3 molecule.

Answer:



Let us consider, the three N-H bond as the basis of NH_3 molecule.

NH_3 molecule belongs to the C_{3v} point group. The complete set of characters of the reducible representation of the NH_3 molecule is,

C_{3v}	E	$2C_3$	$3\sigma_v$
	3	0	1

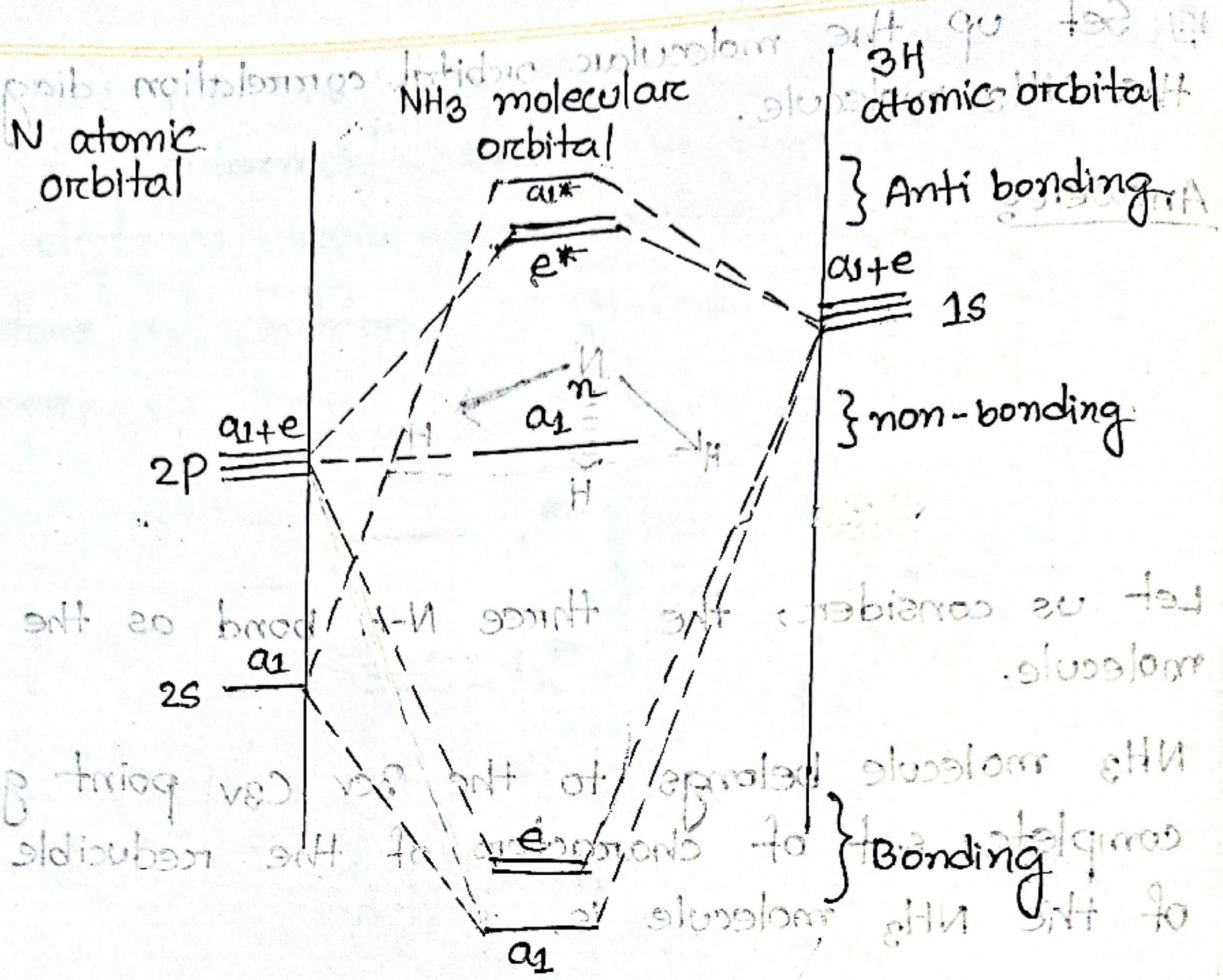
According to the character table of C_{3v} point group,

$A_1 = 1$
$A_2 = 0$
$E = 1$

So, the irreducible representation $= A_1 + E$

s, P_z belongs to A_1 representation
P_x, P_y " " E "

So, the molecular orbital correlation diagram for NH_3 molecule is,



In NH_3, nitrogen has five valance electrons and 3H has three valance electrons. So putting $(5+3)=8$ electrons in the molecular orbitals as follows:

a_1^*, e^* } Antibonding

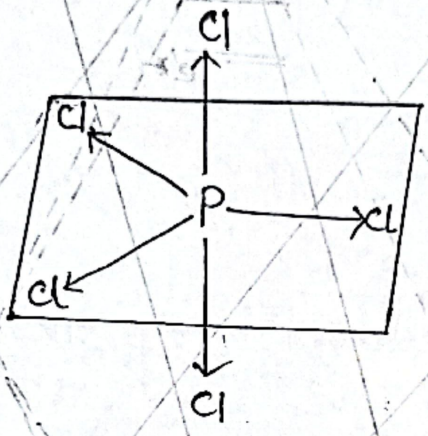
a_1 (n) } Non-bonding

e, a_1 } Bonding

Bond order $= \frac{1}{2}(6-0) = 3$

Set up the molecular orbital correlation diagram for PCl_5 molecules.

Answer:



Let us consider, 5 P–Cl bond as the basis of PCl_5 molecule. PCl_5 belongs to D_{3h} point group. The complete set of character of the reducible representation is,

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	5	2	1	3	0	3

According to the character table of D_{3h} point group,

[calculation]

The irreducible representation is $= 2A_1' + E' + A_2''$

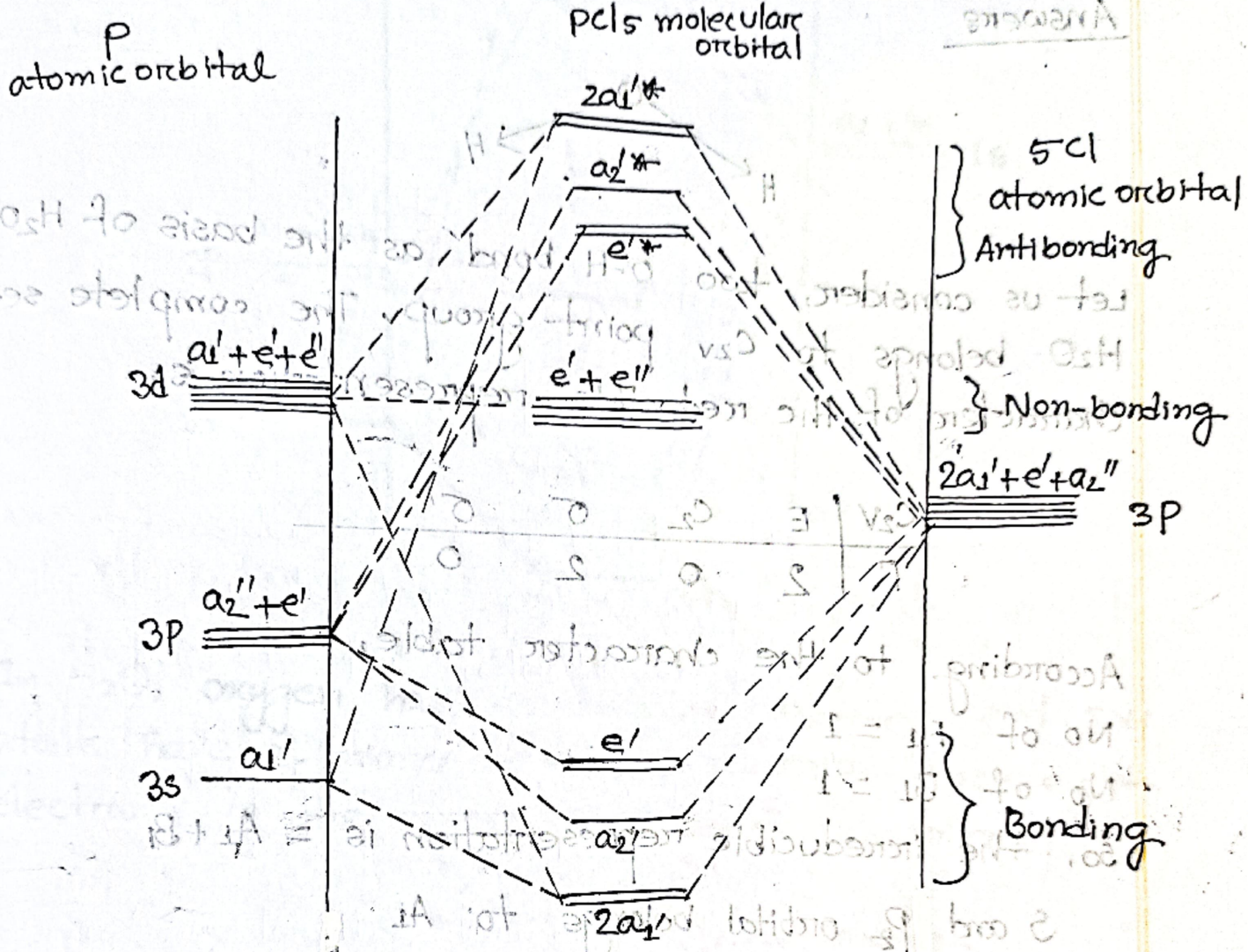
s, dz^2 belongs to A_1'

(P_x, P_y), (dx^2-y^2, dxy) " " E'

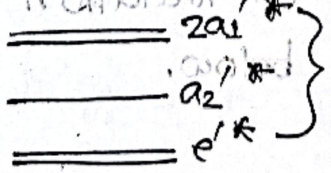
P_z " " A_2''

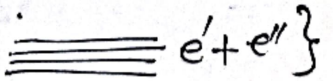
dxz, dyz " " E''

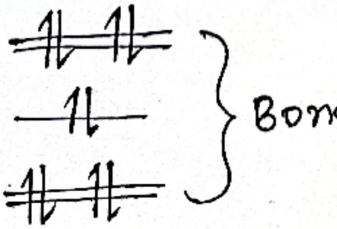
So, the molecular orbital co-relation diagram for PCl_5 molecule is given below:



In PCl_5, there are total 10 electrons in the bonding molecular orbital and has electron in the antibonding.







$\therefore$ Bond order $= \frac{10-0}{2}$

$= 5$

Set the correlation diagram for H_2O molecule.

Answers

Let us consider, two O-H bond as the basis of H_2O molecule. H_2O belongs to C_{2v} point group. The complete set of characters of the reducible representation is;

C_{2V}	E	C_2	σ	σ'
Γ	2	0	2	0

According to the character table,

No of $A_1 = 1$

No of $B_1 = 1$

So, the irreducible representation is $= A_1 + B_1$

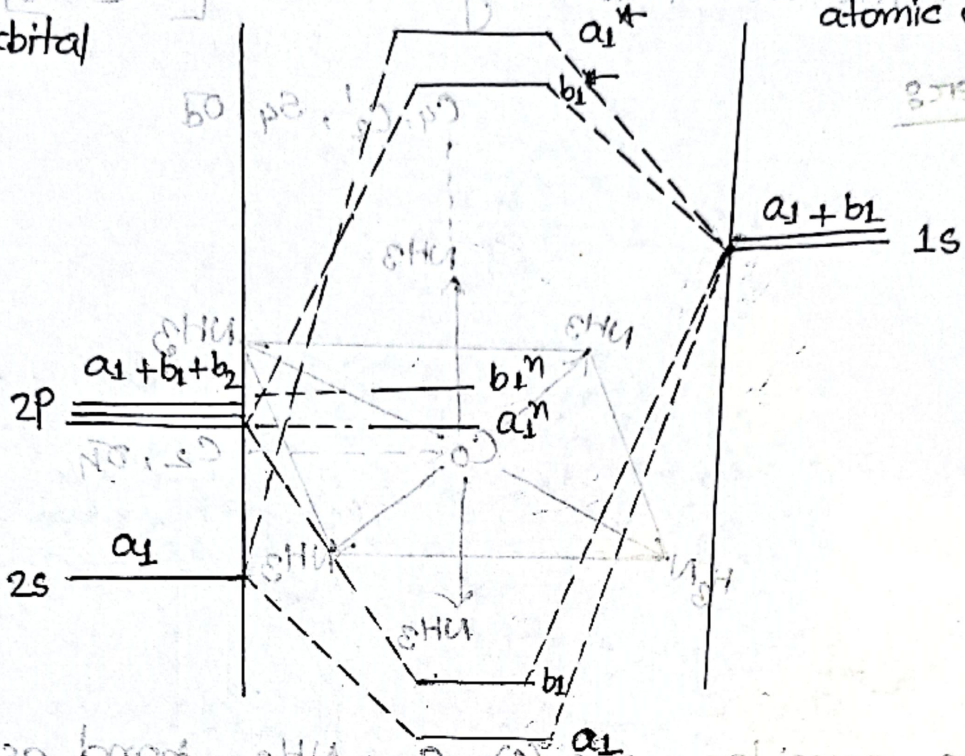
S and P_z orbital belongs to A_1

P_x " " " B_1

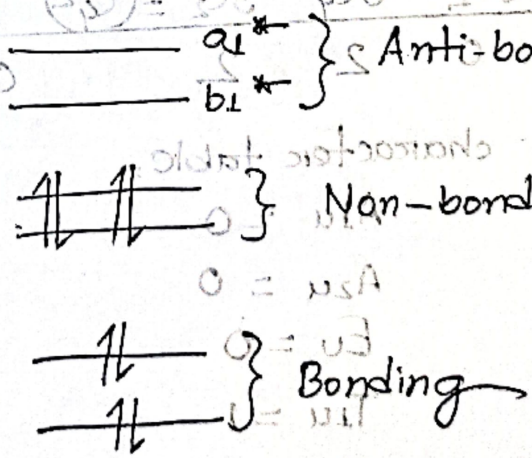
P_y " " " B_2

So, the molecular orbital co-relation diagram for the H_2O molecule is given below.

H_2O M.O.



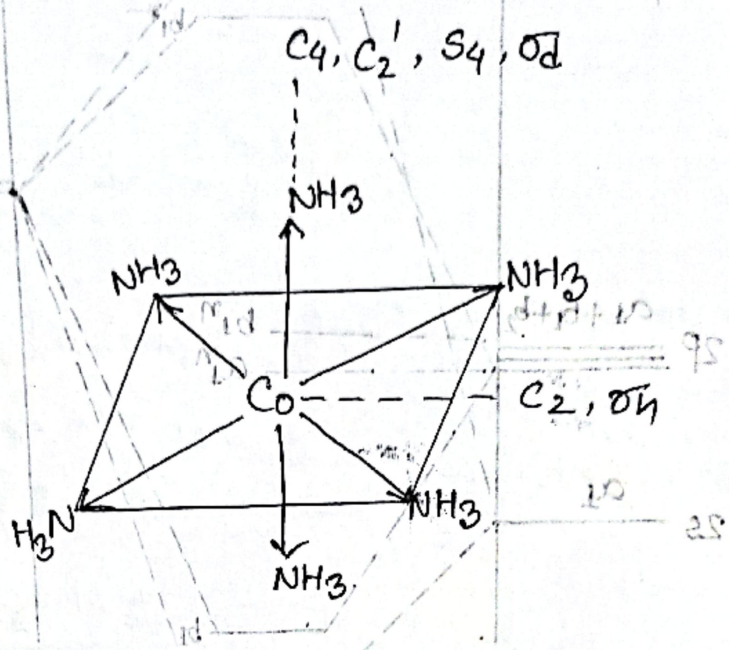
In H_2O, oxygen has six valance electrons and 2-H atoms have 2-valance electron. putting these eight electrons in the molecular orbitals as follows:



$\therefore$ Bond order $= \frac{1}{2}(4-0) = 2$

Set the co-rrelation diagram for $[Co(NH_3)_6]^{3+}$

Answers:



Let us consider six (6) Co - NH_3 bond as the basis of $[Co(NH_3)_6]^{3+}$ molecule/complex. It belongs to Oh point group. The complete set of character of the reducible representation is,

Oh	E	$8C_3$	$6C_2$	$6C_4$	$3C_2' = (C_4^2)$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$
Γ	6	0	0	2	2	0	0	0	4	2

According to character table,

$A_{1g} = 1$ $A_{1u} = 0$

$A_{2g} = 0$ $A_{2u} = 0$

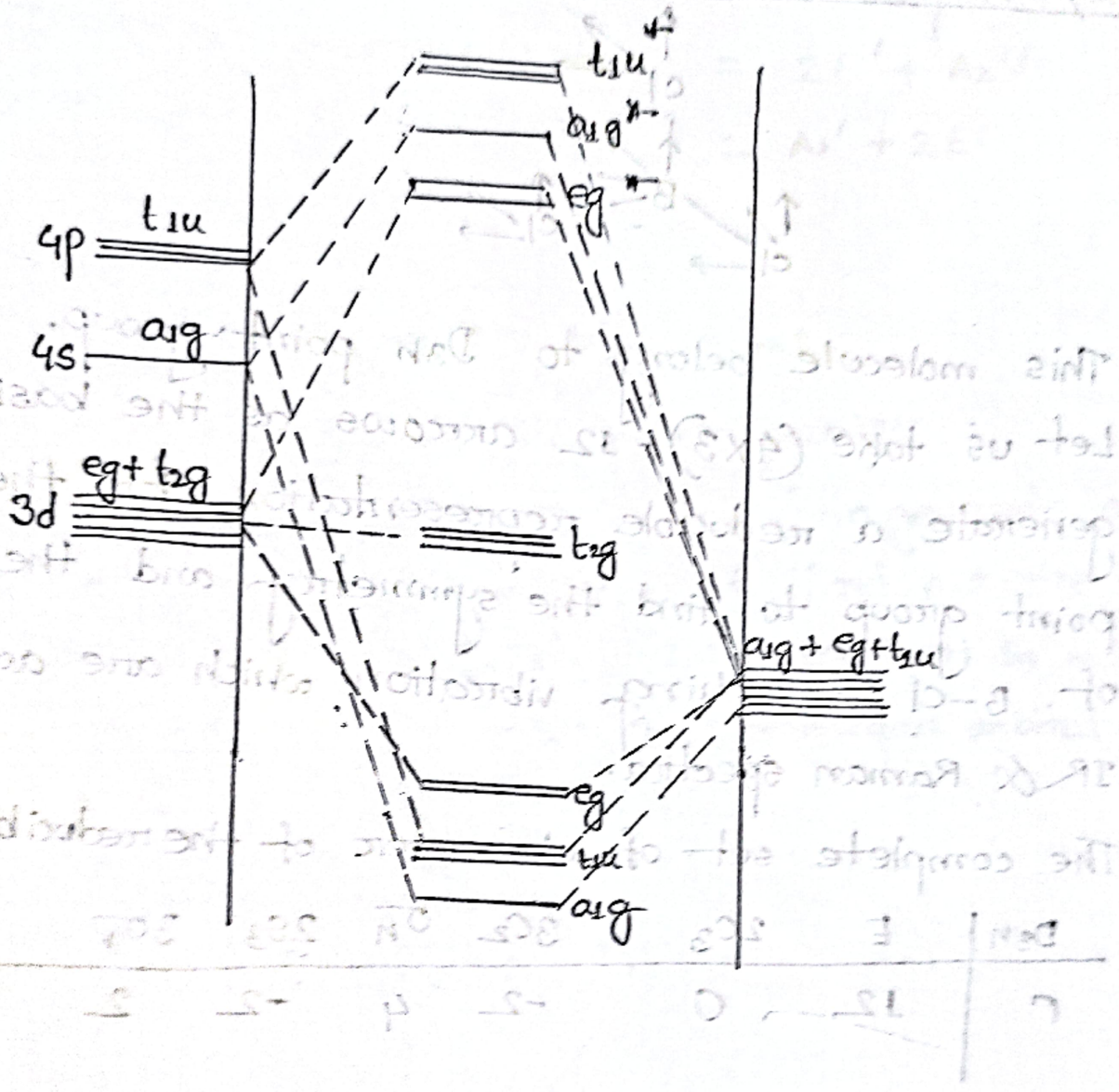
$E_g = 1$ $E_u = 0$

$T_{1g} = 0$ $T_{1u} = 1$

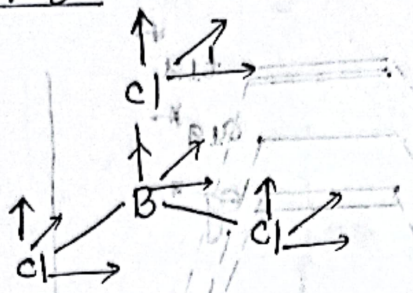
$T_{2g} = 0$ $T_{2u} = 0$

So, the irreducible representation is = $A_{1g} + E_g + T_{1u}$

So, the molecular orbital correlation diagram for the $[Co(NH_3)_6]^{3+}$ complex is given below.


t1u*
a1g*
eg*
4p
t1u
4s
a1g
eg + t2g
3d
t2g
a1g + eg + t1u
eg
t1u
a1g

IR/ Raman active:



This molecule belongs to D_{3h} point group.
Let us take $(4\times3) = 12$ arrows as the basis to generate a reducible representation for the D_{3h} point group to find the symmetry and the number of B–Cl stretching vibration which are active in the IR & Raman spectra.

The complete set of characters of the reducible representation

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
Γ	12	0	-2	4	-2	2

According to the character table of D_{3h} point group:

$A_1' = 1$ $\quad A_2'' = 1$
$A_2' = 1$ $\quad A_1'' = 0$
$E' = 3$ $\quad E'' = 1$

The irreducible representation, $A_1' + A_2' + 3E' + 2A_2'' + E''$
Total molecular movement $= 12$

Total movement $= A_1' + A_2' + 3E' + 2A_2'' + E''$
Rotations $= A_2' + E''$
Translations $= E' + A_2''$

$\therefore$ vibration $= A_1' + 2E' + A_2''$

There are total six mode of vibrations possible.
The IR active vibrations are = $2E' + A_2''$
" Raman " " " = $A_1' + 2E'$

~~Hybrid orbitals~~ Molecular orbital diagrams

This molecule belongs to Td point group.
Let us take four C-H bond as basis to generate a representation for Td point group to find out the s of the group orbitals of the four hydrogen 1s orb We need to know this to draw the molecular orbital d of CH_4 molecule.